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Cloud Native Networking, Security, and Observability

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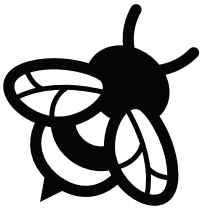
Nico Vibert, Filip Nikolic & James Laverack



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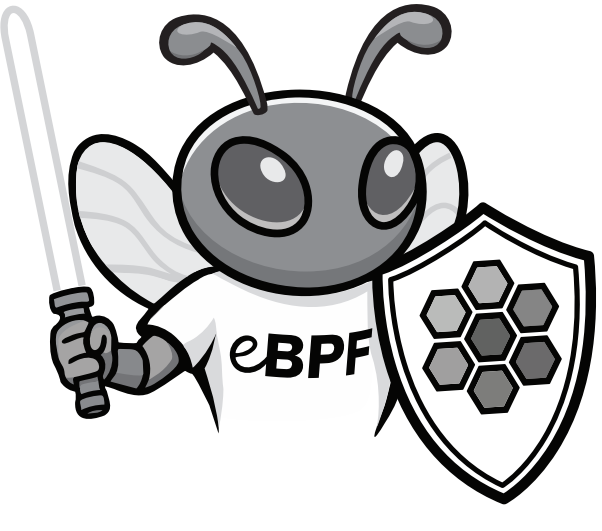

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Nico Vibert, Filip Nikolic, and James Laverack

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by Nico Vibert, Filip Nikolic, and James Laverack

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Brief Table of Contents (*Not Yet Final*)

CHAPTER 1

Inside Cilium

A Note for Early Release Readers

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This will be the 2nd chapter of the final book. Please note that the GitHub repo is available at *https://github.com/isovalent/cilium-up-and-running*.

If you'd like to be actively involved in reviewing and commenting on this draft, please reach out to the editor at *gobrien@oreilly.com*.

Cilium is a powerful eBPF based networking and security solution for Kubernetes and beyond. Understanding its architecture is crucial before you install it and get up and running. This chapter provides an overview of Cilium's core components and how they interact and work together to address the use cases described in Chapter 1. It will provide you with a clear overview of how functionalities are distributed among the various components of Cilium.

By the end of this chapter you will have the knowledge to identify which components are involved in different scenarios, empowering you to approach later sections with confidence.

Cilium at a Glance

You read in Chapter 1 about the cloud native networking, security and observability use cases Cilium addresses. Cilium's versatility can be attributed to its modular

architecture: it is made up of several internal components, each providing different functionalities.

- Cilium Agent
- Cilium CNI Plugin
- Cilium Operator
- Cilium CLI
- Hubble, with its sub-systems (Hubble Server, Hubble Relay, Hubble CLI and Hubble UI)
- eBPF programs
- Envoy Proxy
- DNS Proxy

Here is a brief overview of how all these building blocks fit together.

The Cilium agent is the core component of Cilium: it installs the Cilium CNI plugin, loads the eBPF programs used to forward, filter and observe traffic and monitors the custom resource definitions (CRDs) created by the Cilium operator. When a pod sends traffic, an eBPF program decides whether to forward or drop the traffic and may send it to a proxy for further processing (Envoy for HTTP-related traffic and DNS for DNS-related requests). The Cilium CLI is a binary that primarily assists with managing Cilium. Meanwhile, the Hubble server observes and enriches traffic data, with Hubble relay aggregating it across all cluster nodes. The Hubble UI and CLI provide options to visualize network activity in a user interface and a terminal, respectively.

Cilium Agent

The Cilium agent, often referred to simply as "Cilium" or the "Cilium daemon," is the core component of Cilium. It is responsible for orchestrating several critical tasks across the Kubernetes cluster, such as installing the CNI plugin and loading eBPF programs. To accomplish this, it runs as a DaemonSet on every node with elevated privileges. The Cilium agent plays a pivotal role in ensuring the efficient operation of both the network and security within the cluster, as shown in Figure 2-1. Its primary responsibilities include:

Installing the CNI Plugin
: The agent manages the Container Network Interface (CNI) plugin, ensuring that Pod networking is correctly configured and providing seamless network connectivity.

Loading eBPF Programs and Maps
: The agent loads and manages eBPF (extended Berkeley Packet Filter) programs and maps on each node. These programs enforce network policies, redirect traffic, perform load balancing, and various other fundamental functions.

Reconciling Kubernetes State
: The Cilium agent continuously monitors the desired Kubernetes state by communicating with the kube-apiserver. It checks for changes in Pods, NetworkPolicies and other objects. The agent then updates eBPF maps as needed to reflect these changes.

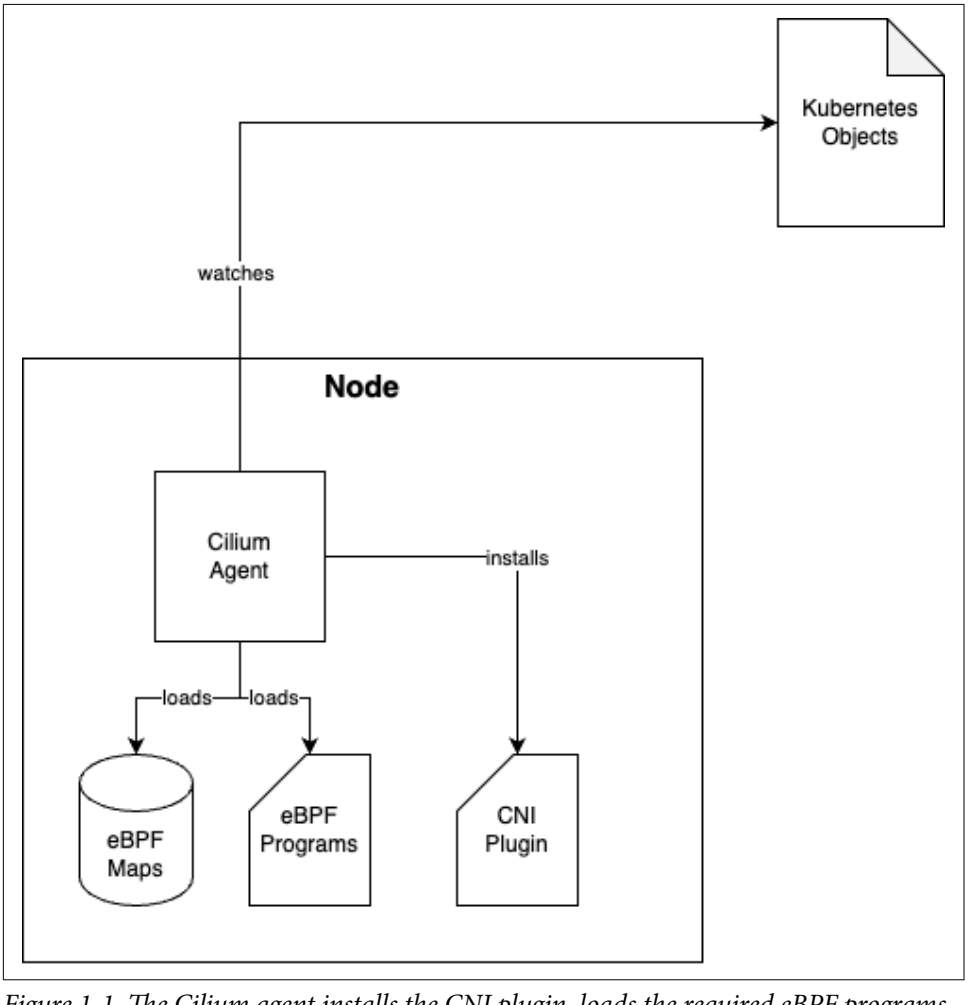

Figure 1-1. The Cilium agent installs the CNI plugin, loads the required eBPF programs and eBPF maps into the kernel and continuously watches over various Kubernetes objects, such as Pods and NetworkPolicies. It then updates the eBPF map accordingly.

For debugging and troubleshooting, the Cilium agent provides a suite of tools when executing into the Cilium agent Pod. The most notable is cilium-dbg. This tool helps gather detailed diagnostic information on the node and should not be confused with the Cilium CLI, which is used for managing Cilium at a higher level across the entire cluster.

Previously, `cilium-dbg` was named `cilium`, which caused confusion since it was unclear whether people were referring to the "Cilium CLI" or the Cilium agent debugging tool. Keep this in mind when reviewing older issues or documentation.

Cilium CNI Plugin

Cilium is often referred to as a CNI (Container Network Interface), but its CNI functionality is just one component of a larger system working together to manage networking in Kubernetes environments.To better understand the role of Cilium CNI let's quickly talk about CNIs in general.

The Container Network Interface (CNI) is a CNCF project that specifies the relationship between a Container Runtime, such as containerd or CRI-O, responsible for container creation, and a CNI plugin tasked with configuring network interfaces within the container upon execution. Ultimately, the CNI plugin performs the substantive tasks of configuring the network, while CNI primarily denotes the interaction framework. However, it's common practice to simply refer to the CNI plugin as "CNI".

Picture a scenario where a user initiates the creation of a Pod and submits the request to the kube-apiserver (through, for example, applying a manifest with `kubectl`). Following the scheduler's determination of the node where the Pod should be deployed, the kube-apiserver lets the corresponding kubelet know. The kubelet, rather than creating containers directly, delegates this task to a container runtime. The container runtime's responsibility encompasses container creation, including the establishment of a network namespace. Once this setup is complete, the container runtime calls upon a CNI plugin to generate and configure virtual ethernet devices and necessary routes.

This flow is depicted in Figure 2-2.

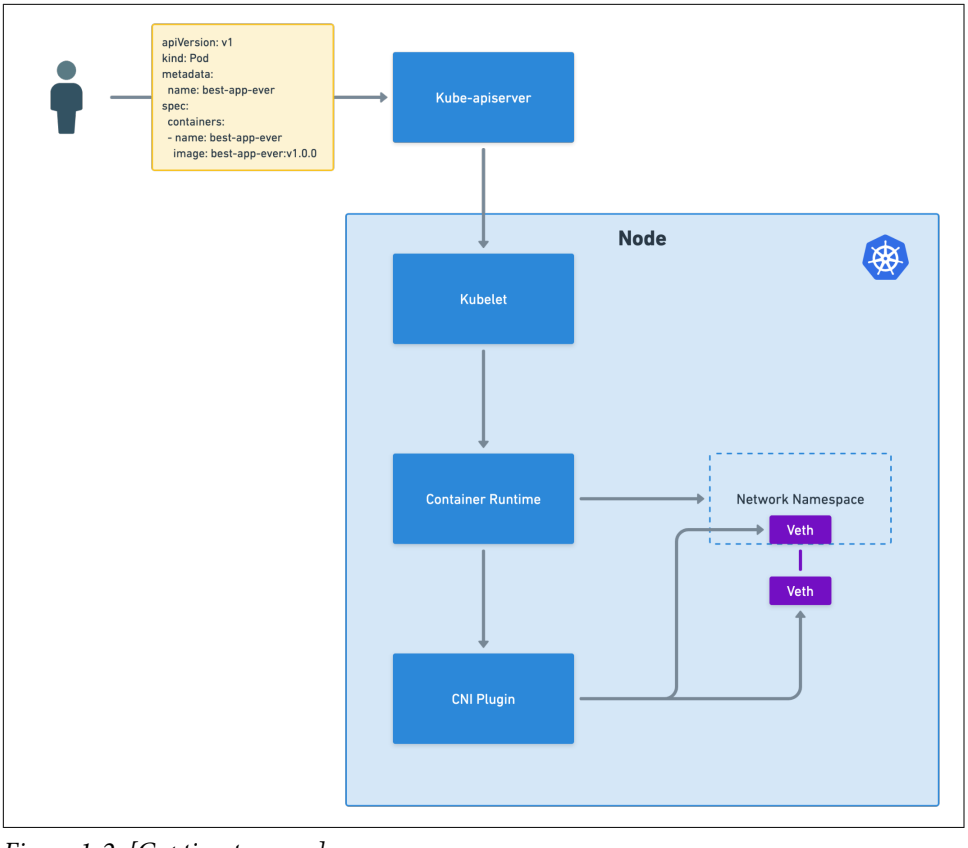

Figure 1-2. [Caption to come]

For more details on the process of a CNI, visit https://github.com/f1ko/demystifying-cni or https://github.com/container-networking/cni

The Cilium CNI consists of two key files located on every node:

- */etc/cni/net.d/05-cilium.conflist*
- */opt/cni/bin/cilium-cni*

The Cilium agent creates these files, which are then read and executed by the container runtime whenever a Pod is created, deleted, or during periodic status checks.

Please note that CNIs typically do not handle traffic forwarding or load balancing. By default, kube-proxy serves as the default network proxy in Kubernetes which utilizes technologies like iptables, IPVS, and more recently nftables, to direct incoming network traffic to the relevant Pods within the cluster. However, Cilium offers a superior alternative by loading eBPF programs into the kernel, achieving the same tasks with significantly better performance, especially at scale. For more information on this topic, see Chapter 5.

Cilium Operator

While the Cilium agent operates at the node level, the Cilium operator functions at the cluster level and runs as a Deployment within the Kubernetes cluster. Certain features and environments may modify the operator's responsibilities, but two key tasks stand out: managing Custom Resource Definitions (CRDs) and handling IP address management (IPAM).

Unlike many other projects, when Cilium is installed using `helm install`, it does not automatically install CRDs. This approach allows the Cilium operator to manage CRDs across versions, ensuring backward compatibility where needed.

This means that Cilium custom resources, such as `CiliumNetworkPolicy`, cannot be applied during the initial installation of Cilium. Before applying these resources, you must wait for the Cilium operator to start up and successfully register the CRDs.

Beyond CRD registration, the Cilium operator also manages IP address allocation (IPAM) as shown in Figure 2-3. By default, it creates a `CiliumNode` object for each node (after registering the CRDs) and populates it with relevant information, such as the podCIDRs. Since the operator operates at the cluster level, it is aware of the IP ranges used by other nodes, therefore ensuring no overlaps. The Cilium agent then reads the data in the `CiliumNode` object to perform its duties.

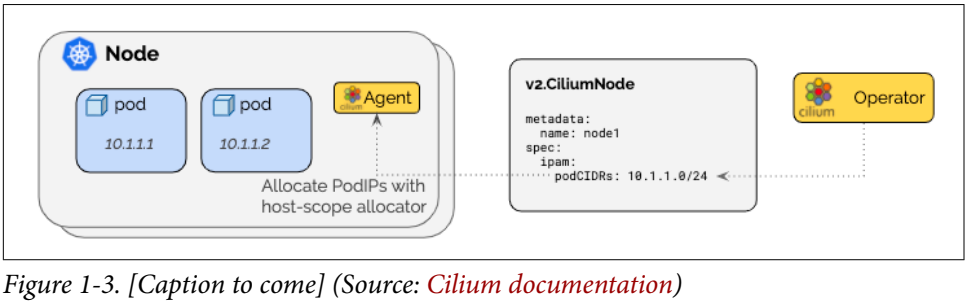

Figure 1-3. [Caption to come] (Source: Cilium documentation)

Cilium CLI

To facilitate the management and configuration of Cilium, a separate command-line binary named Cilium CLI (Command Line Interface) can be installed on your machine.

The Cilium CLI binary enables you to do the following:

- Monitor the status and health of Cilium
- Run a battery of connectivity tests
- Execute operational commands, such as rotating encryption keys and collecting sysdumps
- Install, uninstall Cilium
- Upgrade and downgrade Cilium
- Enable and disable Cilium features

The most commonly used Cilium CLI command is `cilium status`. You can use it to easily see the health of the Cilium components described in this section.

```
$ cilium status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:             OK
 \__/¯¯\__/    Operator:           OK
 /¯¯\__/¯¯\    Envoy DaemonSet:    OK
 \__/¯¯\__/    Hubble Relay:       disabled
    \__/       ClusterMesh:        disabled

DaemonSet              cilium             Desired: 4, Ready: 4/4, Available: 4/4
DaemonSet              cilium-envoy       Desired: 4, Ready: 4/4, Available: 4/4
Deployment             cilium-operator    Desired: 1, Ready: 1/1, Available: 1/1
Containers:            cilium             Running: 4
                       cilium-envoy       Running: 4
                       cilium-operator    Running: 1
Cluster Pods:          3/3 managed by Cilium
Helm chart version:    1.17.1
Image versions         cilium             quay.io/cilium/
cil-
ium:v1.17.1@sha256:8969bfd9c87cbea91e40665f8ebe327268c99d844ca26d7d12165de07f702
866: 4
                       cilium-envoy       quay.io/cilium/cilium-
envoy:v1.31.5-1739264036-958bef243c6c66fcfd73ca319f2eb49fff1eb2ae@sha256:fc708bd
36973d306412b2e50c924cd8333de67e0167802c9b48506f9d772f521: 4
                       cilium-operator    quay.io/cilium/operator-
generic:v1.17.1@sha256:628becaeb3e4742a1c36c4897721092375891b58bae2bfcae48bbf442
0aaee97: 1
```

You will use Cilium CLI throughout the book — you might want to download and install it on your local machine if you intend to follow the examples. The Cilium CLI uses information from the *kubeconfig* to access the cluster via the Kubernetes API. The Cilium CLI is available for Linux, macOS, and Windows machines.

Note that a Cilium CLI client is installed by default inside the Cilium agent container, which goes by the name cilium-dbg. The "in-agent" CLI interacts with the Cilium agent running on the same node and provides more verbose information about the Cilium configuration and state, down to the eBPF maps. In this book, references to Cilium CLI will imply the "client CLI", unless we specify otherwise.

We will provide more information about the "in-agent CLI" in Chapter 16: Operations.

Hubble

Hubble, a sub-system of Cilium, provides network observability. Hubble builds on top of Cilium and eBPF to provide deep network visibility information of Kubernetes applications and services by tapping into Cilium's eBPF datapath. Hubble is an optional component of Cilium and is disabled by default.

Hubble is made up of several components (Figure 2-4). The Hubble server captures data locally, the relay aggregates it across the cluster while the UI and the CLI provide two options to interact and consume the data aggregated by the Relay.

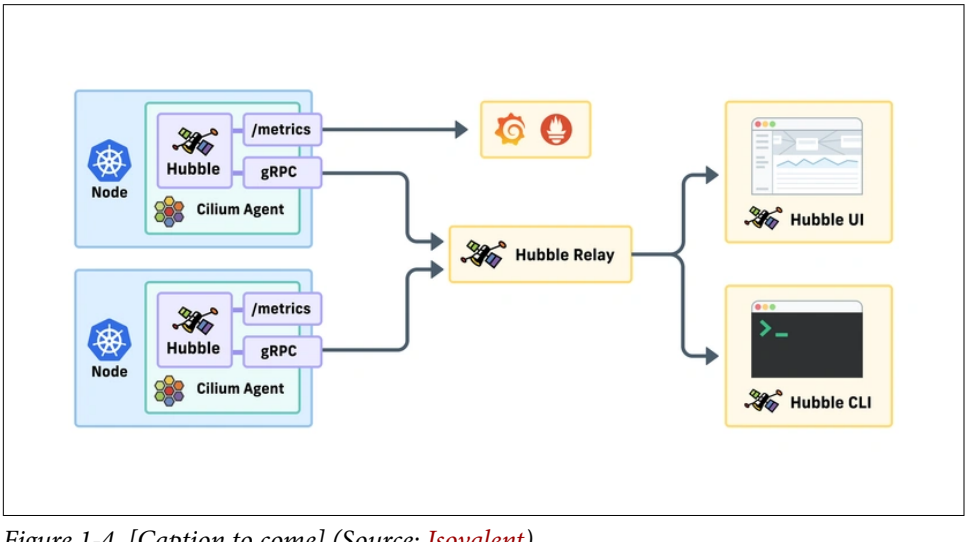

Figure 1-4. [Caption to come] (Source: Isovalent)

Hubble Server

The Hubble server runs on each node and retrieves the eBPF-based visibility from Cilium. The server is embedded into the Cilium agent and captures *flow logs*, which are then gathered across all cluster nodes by the Hubble relay to provide a richer cluster-wide view of network traffic. The flow logs can then be displayed in the Hubble UI or interpreted in the Hubble CLI for monitoring, troubleshooting and alerting.

A flow log includes a wealth of information about:

- a timestamp
- a source pod, along with its namespace, port, and Cilium identity
- the direction of the flow (->, <-, or at times <> if the direction could not be determined)
- a destination pod, along with its namespace, port, and Cilium identity
- a trace observation point (e.g. to-endpoint, to-stack, to-overlay)
- a policy verdict (e.g. FORWARDED or DROPPED)
- a layer 4 protocol (e.g. UDP, TCP), optionally with flags

Don't worry if you are not familiar with concepts such as policy verdict and identities, we will cover them in Chapter 12 (Network Policy).

Here is an example of a flow log for an ICMPv6 packet sent from a pod named `pod-worker` on a node named `kind-worker` to a pod named `pod-worker2` located on node `kind-worker2`.

```
$ hubble observe --ipv6 --from-pod pod-worker
Sep  7 15:11:18.288: default/pod-worker (ID:3211) -> default/pod-worker2
(ID:50760) to-overlay FORWARDED (ICMPv6 EchoRequest)
Sep  7 15:11:18.289: default/pod-worker (ID:3211) -> default/pod-worker2
(ID:50760) to-endpoint FORWARDED (ICMPv6 EchoRequest)
```

You will explore how to use, interpret, and display flow logs in more detail in Chapter 15.

In order for clients to consume flow logs, the Hubble server exposes a gRPC API over a local UNIX domain socket or via an API, for external systems such as Hubble Relay. The Hubble Server also exposes Prometheus metrics, which can be exported and visualized on dashboards such as Grafana.

Hubble Relay

The Hubble Relay system provides a cluster-wide API for querying Hubble flow data, which can be accessed directly or via the Hubble CLI and UI.

The Hubble Relay consumes the gRPC API from each Hubble Server to provide multi-node support. Hubble Relay connects to every node in a cluster and maintains a persistent connection with every Hubble server across the cluster. Before the Hubble Relay was released, Hubble could only operate on a per-node basis and building a cluster-wide view required a tedious iteration across multiple Hubble instances in the cluster. Hubble Relay aggregates data from all Hubble servers and serves it on one query endpoint.

Given that Hubble Relay collects cluster-wide flows you should treat it as a sensitive component. Connections between Hubble instances and Hubble Relay are secured using mutual TLS by default and users can choose to use tools such as cert-manager to generate the certificates automatically.

When the Hubble feature is enabled, Hubble Relay is automatically deployed on the Kubernetes cluster through a Kubernetes Deployment.

Hubble CLI

The Hubble CLI is a binary you can use to visualize and filter network flow logs collected from the Hubble Relay.

Just like Cilium CLI, Hubble CLI is installed and managed separately from Cilium and Hubble. Platform engineers would typically install Hubble CLI on their machine and run Hubble CLI for forensics and troubleshooting purposes. Hubble CLI connects to the Hubble Relay component to provide a cluster-wide view of traffic.

```
$ hubble observe --to-fqdn api.github.com
Aug  3 15:12:13.929: tenant-jobs/crawler-5c645d68f4-qchk8:47180 (ID:21067) ->
api.github.com:80 (world) policy-verdict:all EGRESS ALLOWED (TCP Flags: SYN)
```

Note that Hubble CLI can be used *within* the Cilium agent to display flow logs captured by Hubble. In that case, the flow logs reported would only be the logs observed by the local Cilium agent, rather than the cluster-wide logs aggregated by Hubble Relay.

The Hubble CLI binary is installed by default in Cilium agent pods. With the following kube-ctl command, you can observe traffic locally to view the last observed flow on the node where the Cilium agent is deployed:

```
$ kubectl exec -n kube-system -it cilium-gb945 -c cilium-agent -- hubble
observe --last 1
Feb 20 12:18:35.872: 10.244.2.7:58762 (host) ->
kube-system/hubble-relay-d9495cdc-wfd6l:4222 (ID:3873) to-endpoint FORWARDED
(TCP Flags: ACK)
```

The Hubble CLI is available for Linux, macOS, and Windows machines.

Hubble UI

The other method to consume the data collected by Hubble is through the Hubble User Interface. The graphical UI utilizes the aggregated data from Hubble Relay to provide a graphical service dependency and connectivity map.

Hubble UI is not installed by default when you install Hubble. It must be installed separately.

Hubble UI can also visualize all flows across the cluster. The information displayed is namespace-based. By selecting the `kube-system` namespace, you can visualize the interactions between Hubble UI and Hubble Relay itself.

eBPF Programs & eBPF Maps

eBPF (extended Berkeley Packet Filter) is a revolutionary technology that enables users to extend the behavior of the Linux kernel without the need to submit patches, wait for review, or rebuild and update the kernel. Instead, eBPF allows for the creation of custom code that can be directly injected into the kernel, providing immediate changes to its behavior—without requiring a reboot.

One of the primary reasons eBPF has gained widespread adoption is its ability to provide safe execution within the kernel. Furthermore, it enables users to write highly efficient programs that execute at precise points in the kernel's workflow. By attaching programs to the most relevant hooks in the kernel, users can act at the earliest possible moment. This improves performance by avoiding unnecessary work later in the kernel's execution cycle.

Two key concepts are fundamental to understanding eBPF:

eBPF programs
: Stateless, event-driven pieces of code triggered by specific events within the kernel. These programs are attached to "hook points" in the kernel and when those points are reached, the corresponding eBPF program is executed. Therefore, the programs can operate in real-time, providing immediate response to kernel events.

eBPF maps
: Serve as in-kernel data storage. These maps allow eBPF programs (and other processes) to store and retrieve data, enabling dynamic decision-making. Just as a stateless microservice may need to query a database for information before making a decision, eBPF programs can perform lookups in eBPF maps to guide their actions based on the stored data.

In the context of Cilium, eBPF programs are used to implement critical networking and security functions (Figure 2-5). For example, eBPF programs can be attached to the kernel's network-relevant hook points to inspect network packets and determine if they should be allowed or dropped by Network Policies. To help with the decision, the eBPF program performs a lookup in an eBPF map containing information about allowed sources and destinations. The Cilium agent continually updates these eBPF maps to reflect the desired state as specified by NetworkPolicies.

Not all network or security decisions in Cilium are made using eBPF. Although eBPF has the potential to handle more complex decisions, such as layer 7 policies, this is not the way Cilium is implemented. At the time of writing, layer 7 decisions are handled in userspace, typically by an Envoy or DNS proxy.

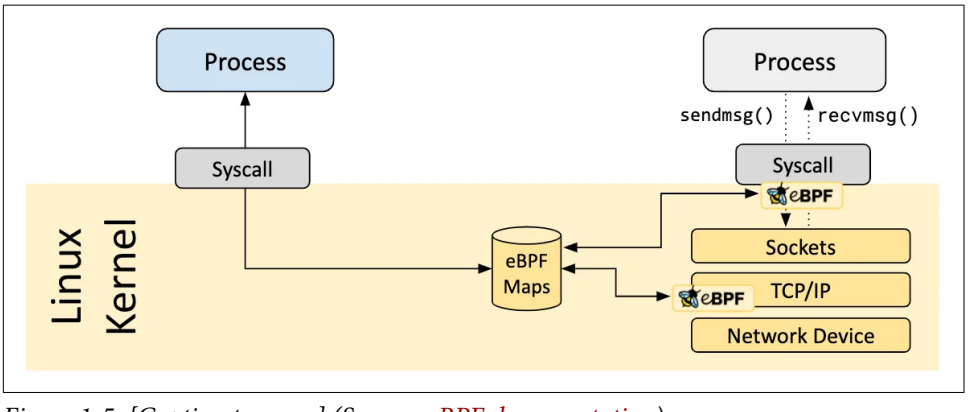

Figure 1-5. [Caption to come] (Source: eBPF documentation)

While eBPF was initially developed for the Linux kernel, its potential has led to efforts to port eBPF to other kernels, such as the Windows kernel, broadening its use across different platforms.

Envoy Proxy

Envoy is a high-performance open source proxy and a key project within the CNCF, widely recognized for its scalability and robust feature set. In Cilium, Envoy plays a crucial role in managing and controlling HTTP traffic, enforcing HTTP-based policies and supporting sophisticated traffic management decisions.

Envoy implements important features such as HTTP-based CiliumNetworkPolicies, Ingress & Gateway API, and more. These features allow for fine-grained control over HTTP traffic, ensuring that routing decisions, security policies, and other configura-

tions can be applied based on specific HTTP headers, methods, paths, and additional data contained within the HTTP payload.

To accomplish this, HTTP traffic is routed through the Envoy proxy where layer 7 interactions are needed.

To minimize performance penalties Envoy runs on each node through its own dedicated DaemonSet, as you can see in the following code:

```
$ kubectl get -n kube-system daemonset/cilium-envoy
NAME         DESIRED CURRENT READY UP-TO-DATE AVAILABLE NODE SELECTOR
AGE
cilium-envoy 3       3       3     3          3         kubernetes.io/os=linux
28h
```

Sending traffic through Envoy ensures that security and traffic management can be handled in a highly dynamic environment, particularly for cloud-native applications. By offloading HTTP-related decisions to Envoy, Cilium is able to leverage Envoy's extensive capabilities for managing HTTP traffic, all while continuing to rely on eBPF-based networking decisions for lower layers of the stack.

Since HTTP functionality relies on the Envoy proxy, applications will be impacted if the Envoy proxy becomes unavailable.

DNS Proxy

The DNS Proxy in Cilium enables network security decisions to be made based on fully qualified domain names (FQDNs), adding an important layer of granularity to security enforcement. When using FQDN-based network policies, the DNS proxy listens to and captures DNS requests and responses, allowing it to learn which domain names resolve to which IP addresses. Once this mapping is established, the network policies can allow or deny traffic based on domain names To achieve this, it's essential that DNS traffic is routed through the DNS proxy.

The DNS proxy is especially beneficial in dynamic, cloud-native environments where IP addresses are often ephemeral and subject to change. In these scenarios, services might scale or change their IP addresses frequently. The DNS proxy ensures that security policies remain effective in such environments by enforcing rules based on domain names rather than static IPs. For example, blocking traffic to a specific domain becomes possible, regardless of the IP addresses that may shift over time, providing a more flexible approach to network security.

The DNS proxy also enhances Hubble flow data with DNS service identities, enabling you to monitor and troubleshoot DNS activities and identify anomalous behaviour. Once enabled, Hubble will report the names of services in its interface.

Although the DNS proxy is considered a separate component in Cilium, it operates on every node within the cluster as part of the Cilium agent Pod. This approach minimizes performance overhead by ensuring that DNS traffic is routed locally on the same node, reducing the need for inter-node communication.

Since FQDN functionality relies on the DNS proxy, which is part of the Cilium agent, applications will be impacted if the Cilium agent becomes unavailable.

You will learn more about the DNS proxy in Chapter 12 (Network Policy).

Another optional component of Cilium is the Cluster Mesh API server.

Cilium Cluster Mesh provides connectivity and service load balancing across multiple clusters and leverages a component called the Cluster Mesh API server. You will learn more about it in Chapter 8.

Putting it all together

In this chapter we have explored the individual components of Cilium. Let's dive into how they interact with each other to get a better understanding of the big picture (Figure 2-6).

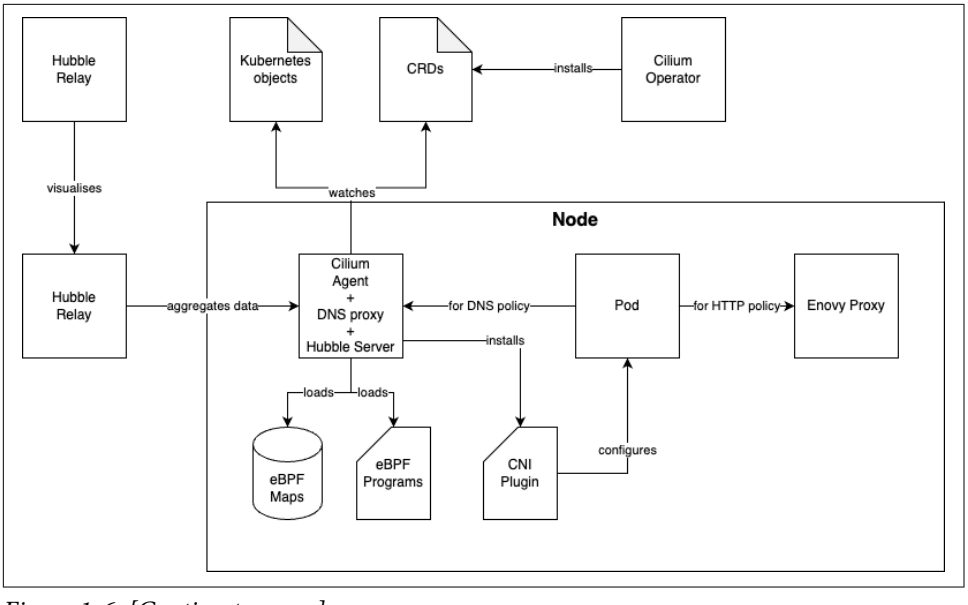

Figure 1-6. [Caption to come]

The Cilium Operator is responsible for creating various CRDs. These CRDs, along with other Kubernetes objects, are monitored by the Cilium agent. The Cilium agent uses this information to load and configure eBPF maps and programs as needed. Additionally, the agent installs the CNI plugin, which is responsible for setting up networking for each new Pod.

When a Pod sends traffic it reaches a hook point where an eBPF program is triggered. This program evaluates the traffic and decides whether to allow or drop it. If layer 7 or FQDN policies are in use, the eBPF program forwards the traffic to either the DNS proxy for DNS-related requests or the Envoy proxy for HTTP-related traffic.

The Envoy proxy processes the traffic and applies its own logic to decide whether to allow or drop it. For DNS traffic, the DNS proxy captures DNS responses, maps IP addresses to FQDNs and provides this information to the Cilium agent. The agent then updates the eBPF maps to allow traffic to the IPs associated with specific FQDNs.

Meanwhile, the Hubble server observes all traffic on the node. It also uses data from the DNS proxy to enrich the traffic observations, providing not just IP addresses but also their corresponding domain names. Hubble Relay aggregates traffic data from all the Hubble servers running across nodes, enabling a centralized view of network traffic within the cluster. Finally, the Hubble UI queries Hubble relay and visualizes the aggregated traffic data, offering an intuitive interface for exploring and analyzing network activity within the cluster.

With that, we've established a solid understanding of the various components, allowing us to explore each one in greater depth in the following chapters.

CHAPTER 2
Getting Started with Cilium

A Note for Early Release Readers

With Early Release ebooks, you get books in their earliest form—the author's raw and unedited content as they write—so you can take advantage of these technologies long before the official release of these titles.

This will be the 3rd chapter of the final book. Please note that the GitHub repo is available at *https://github.com/isovalent/cilium-up-and-running*.

If you'd like to be actively involved in reviewing and commenting on this draft, please reach out to the editor at *gobrien@oreilly.com*.

Now that you're familiar with the core components of Cilium, let's go ahead and deploy it. The best way to become familiar with Cilium is to deploy in a Kubernetes cluster and try out some of the core capabilities. In this section, you will learn how to:

- Install Cilium in a Kubernetes cluster
- Deploy an application,
- Access, secure and monitor the application.

In later chapters, we will dive further into various areas of Cilium, but a practical hands-on experience with Cilium will give you strong foundational knowledge for the rest of the book. To get started, you will first need access to a Kubernetes cluster. We'll create one locally.

If you don't have local compute resources available to run a Kubernetes cluster, feel free to use the free online hands-on labs provided by Isovalent, the creators of Cilium. Head over to isovalent.com/labs to begin your evaluation of Cilium.

Setting Up Your Environment

Throughout this book, you will find code snippets and scripts to install, test and configure Cilium. This implies that you have access to a Kubernetes cluster. Do not try any of the scripts in a production cluster: they may cause some disruption or even worse, a complete cluster outage.

Cilium works with any distribution conformant with "vanilla" Kubernetes and is often even packaged and included by default with many Kubernetes distributions and services.

Not everyone has the financial or computing resources to afford a Kubernetes cluster. Therefore, the environment we will recommend for the vast majority of the exercises in the book is based on Kubernetes in Docker (kind), a popular tool used to run Kubernetes clusters locally in Docker containers.

We recommend kind over alternatives because:

- It works across multiple platforms (Linux, macOS and Windows)
- It is a CNCF-certified conformant Kubernetes installer
- It's lightweight compared to some of the alternatives.

That said, if you prefer another lightweight Kubernetes solution, check the documentation to make sure it's compatible with Cilium.

During the creation of this book, we tested all scripts across a variety of Kubernetes environments, including:

- Minikube
- K3S
- Azure Kubernetes Services (Bring your own CNI)
- Google Kubernetes Engine
- Amazon Elastic Kubernetes Services

The majority of features work seamlessly but you can find some considerations and caveats in the book's GitHub repository.

Kubernetes-in-Docker (kind)

This chapter assumes you have Docker or another container runtime installed, as `kind` requires it to launch Kubernetes clusters. If you need help installing Docker, refer to the official documentation for your operating system.

If you don't have `kind` installed yet, let's do that now. For those of you on MacOS machines, you can simply use Homebrew to install it:

```
$ brew install kind
```

The instructions for other platforms can be found on the kind website.

Installing the Cilium CLI

To manage and inspect Cilium deployments, we will use the Cilium CLI. It should be available in your operating system's package manager; for example, you can run the following Homebrew command to install it on MacOs:

```
$ brew install cilium-cli
```

You can also download and install the appropriate tarball from the Cilium CLI GitHub repository (*https://github.com/cilium/cilium-cli/*). Detailed instructions can also be found on the Cilium Docs.

Once installed, verify it has been installed correctly:

```
$ cilium version --client
cilium-cli: v0.18.0 compiled with go1.24.0 on darwin/arm64
cilium image (default): v1.17.0
cilium image (stable): v1.17.3
```

Installing the Hubble CLI

To observe network traffic in the cluster, we will use Hubble and its binary client, the Hubble CLI. Likewise, you can also install it with a package manager like Homebrew:

```
$ brew install hubble
```

Alternatively, download and install the tarball from the Hubble GitHub repository (*https://github.com/cilium/hubble*). You can also find detailed installation instructions on the Cilium docs (*https://docs.cilium.io/en/stable/observability/hubble/setup/#hubble-cli-install*).

Once installed, verify the Hubble CLI has been installed correctly by running the following command.

```
$ hubble version
hubble v1.16.4@HEAD-4b765dc compiled with go1.23.3 on darwin/arm64
```

Installing Helm

We will use the Kubernetes package manager Helm throughout the book. If you don't have it installed on your machine yet, install it via your package manager (e.g. `brew install helm`) or refer to the official Helm installation docs.

Deploying a Kind Cluster

The following YAML manifest, available in the book's GitHub repository, includes the specification of the cluster.

```
kind: Cluster                               # Declares a Kind cluster
apiVersion: kind.x-k8s.io/v1alpha4          # Kind API version
nodes:
- role: control-plane                       # One control-plane node
- role: worker                              # Two worker nodes
- role: worker                              # Two worker nodes
networking:
  disableDefaultCNI: true        # Disables kindnetd prior to Cilium install
```

The most notable aspect of the configuration is the `networking.disableDefaultCNI: true` setting. It specifies that the cluster must be deployed without kind's builtin CNI (kindnetd).

As you recall from Chapter 1, a CNI plugin is required in order for our cluster to host and connect applications. We intentionally disable kind's default built-in CNI so that we can install Cilium instead in the next section.

You can find all the YAML manifests you will use in this chapter to deploy and test Cilium in the chapter03 directory of the book's GitHub repository *https://github.com/isovalent/cilium-up-and-running* . Navigate to this folder before proceeding.

Let's now deploy the kind cluster, with the following command:

```
$ kind create cluster --config kind-cluster-config.yaml
Creating cluster "kind" ...
   Ensuring node image (kindest/node:v1.32.0)     Preparing nodes
   Writing configuration      Starting control-plane   °   Installing Storage-
Class      Joining worker nodes   Set kubectl context to "kind-kind"
You can now use your cluster with:

kubectl cluster-info --context kind-kind

Not sure what to do next?    Check out https://kind.sigs.k8s.io/docs/user/quick-
start/
```

With the cluster now deployed, let's use `kubectl` to check our cluster.

```
$ kubectl get nodes
NAME                 STATUS     ROLES           AGE   VERSION
kind-control-plane   NotReady   control-plane   46m   v1.32.0
kind-worker          NotReady   <none>          46m   v1.32.0
kind-worker2         NotReady   <none>          46m   v1.32.0
```

The preceding terminal output shows that the four nodes (one control-plane and three workers) have been deployed but they are in `NotReady` status. This indicates the nodes are unhealthy and will not accept workloads. You may have already guessed the reason but let's confirm why by inspecting the status of one of our worker nodes.

```
$ kubectl get nodes kind-worker -o yaml
[...]
- lastHeartbeatTime: "2025-02-21T18:34:29Z"
  lastTransitionTime: "2025-02-21T17:47:48Z"
  message: 'container runtime network not ready: NetworkReady=false reason:Net-
workPluginNotReady message:Network plugin returns error: cni plugin not initia-
lized'
  reason: KubeletNotReady
  status: "False"
  type: Ready
```

As you can see, since no network plugin has been deployed, the Kubelet reports it is not ready to run pods. We'll resolve that by installing the Cilium CNI plugin in the next section.

Installing Cilium

Cilium supports multiple installation methods. The two most common are:

- Using the **Cilium CLI** (`cilium install`) for quick setup and environment auto-detection
- Using **Helm** directly, which provides explicit control and aligns better with GitOps and long-term operations

Let's review both options.

Installing Cilium Using the Cilium CLI

The Cilium CLI simplifies the initial experience by inspecting your current Kubernetes context and generating a set of recommended Helm values based on your environment.

To preview the configuration without performing the installation:

```
$ cilium install --dry-run-helm-values
```

On your kind cluster, this should return:

```
cluster:
  name: kind-kind
ipam:
  mode: kubernetes
operator:
  replicas: 1
routingMode: tunnel
tunnelProtocol: vxlan
```

In this configuration:

- The Cilium operator will be deployed with a single replica
- IPAM mode is set to Kubernetes-host Scope mode
- Routing mode is set to encapsulation, with VXLAN tunnels used to encapsulate traffic between nodes

You will explore IPAM and routing modes in detail in Chapters 4 and 5. For now, note that:

- Cilium will form VXLAN tunnels between all nodes to enable cross-node pod communication
- It will allocate pod IPs from the subnets assigned to each node by Kubernetes

To verify the pod CIDRs Kubernetes has assigned:

```
$ kubectl get nodes kind-worker -o yaml | yq .spec.podCIDR
10.244.1.0/24

$ kubectl get nodes kind-worker2 -o yaml | yq .spec.podCIDR
10.244.2.0/24
```

When you deploy the sample app shortly, you will observe that Cilium will assign IPs from these ranges.

Running `cilium install` now would install Cilium using the auto-generated settings:

```
$ cilium install
  uto-detected Kubernetes kind: kind
   Using Cilium version 1.17.4
  uto-detected cluster name: kind-kind
  uto-detected kube-proxy has been installed
```

Installing Cilium Using Helm (Recommended)

Helm is already widely used to manage other Kubernetes applications, such as Prometheus, Grafana, cert-manager, etc.. Using it for Cilium as well ensures a consistent deployment workflow and makes it easier to integrate with CI/CD and GitOps practices.

To begin, save the auto-generated values into `helm-values.yaml` file:

```
$ cilium install --dry-run-helm-values >> helm-values.yaml
```

Then install Cilium with Helm:

```
$ helm install cilium cilium/cilium -n kube-system --values helm-values.yaml
```

Sample output:

```
NAME: cilium
LAST DEPLOYED: Fri Mar 21 09:39:20 2025
NAMESPACE: kube-system
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
You have successfully installed Cilium with Hubble.

Your release version is 1.17.1.

For any further help, visit https://docs.cilium.io/en/v1.17/gettinghelp
```

Throughout the rest of the book, you'll find Helm values files in the folder corresponding to each chapter. Use them with the Helm install command shown earlier to deploy Cilium with the appropriate settings.

Instead of using a values file, you can pass values inline:

```
$ helm install cilium cilium/cilium -n kube-system \
  --set cluster.name=kind-kind \
  --set ipam.mode=kubernetes \
  --set operator.replicas=1 \
  --set routingMode=tunnel \
  --set tunnelProtocol=vxlan
```

The `cilium install` command also accepts these Helm-style `--set` flags for customization:

```
$ cilium install \
  --set cluster.name=kind-kind \
  --set ipam.mode=kubernetes \
  --set operator.replicas=1 \
  --set routingMode=tunnel \
  --set tunnelProtocol=vxlan
```

By default, both `cilium install` and Helm will install the latest stable version of Cilium. However, you can explicitly specify a version to ensure consistency across environments or to avoid unexpected changes. This is especially useful in production, where you may want to pin versions and manage upgrades intentionally.

To install a specific version, use the following flag:

```
$ helm install cilium cilium/cilium -n kube-system --version 1.17.4 --values
helm-values.yaml
```

After a couple of minutes, Cilium should be successfully installed. Verify the installation with the `cilium status` command.

```
$ cilium status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:             OK
 \__/¯¯\__/    Operator:           OK
 /¯¯\__/¯¯\    Envoy DaemonSet:    OK
 \__/¯¯\__/    Hubble Relay:       disabled
    \__/       ClusterMesh:        disabled

DaemonSet              cilium                   Desired: 3, Ready: 3/3, Avail-
able: 3/3
DaemonSet              cilium-envoy             Desired: 3, Ready: 3/3, Avail-
able: 3/3
Deployment             cilium-operator          Desired: 1, Ready: 1/1, Avail-
able: 1/1
Containers:            cilium                   Running: 3
                       cilium-envoy             Running: 3
                       cilium-operator          Running: 1
                       clustermesh-apiserver
                       hubble-relay
Cluster Pods:          3/3 managed by Cilium
[...]
```

Note that you can watch the progress of the installation by typing `cilium status –wait` and observe the progression of the installation.

In the output you can see how the components we describe in Chapter 2 are deployed. As the Cilium Agent and Cilium Envoy components are both deployed through a DaemonSet, there is an instance of each deployed on every cluster node. As you might have seen in the earlier `cilium install --dry-run-helm-values`, a single Cilium Operator was deployed as part of the deployment.

You can verify with the following `kubectl` commands:

```
$ kubectl get -n kube-system daemonset/cilium
NAME     DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELEC-
TOR            AGE
cilium   3         3         3       3            3           kuber-
netes.io/os=linux   41h
$ kubectl get -n kube-system daemonset/cilium-envoy
NAME           DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELEC-
TOR            AGE
cilium-envoy   3         3         3       3            3           kuber-
```

```
netes.io/os=linux   41h
$ kubectl get -n kube-system deployment/cilium-operator
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
cilium-operator   1/1     1            1           41h
```

Deploying a Sample Application

Let's now deploy a sample application. The following example is a common first sample application deployed in Kubernetes: the lightweight `NGINX` web server app. We'll use the *nginx-deployment.yaml* manifest file from the book's GitHub repository:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      nodeSelector:
        kubernetes.io/hostname: kind-worker2
      containers:
        - name: nginx
          image: nginx
```

"Pinning" the nginx-server pod to a specific worker node (`kind-worker2`) isn't a best practice in production but it's useful here to illustrate inter-node traffic, which Cilium handles through VXLAN tunneling in the selected configuration mode.

Let's deploy the manifest:

```
$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

Next, let's verify that the pod was scheduled on the expected node and that it was assigned an IP address by running the command:

```
$ kubectl get pods -o wide
NAME                               READY   STATUS    RESTARTS   AGE
IP             NODE
nginx-deployment-979f5455f-tjd2w   1/1     Running   0          7s
10.244.2.127   kind-worker2
```

Let's now deploy a pod on a different worker node (kind-worker)using the following manifest. We are using a networking utility called netshoot, which we'll use throughout the book because it includes many helpful networking tools. Given that it's a

debugging utility, it's meant to run once and exit unless we specify to keep it running (which is what we will do, by instructing it to run `sleep infinity`).

```
apiVersion: v1
kind: Pod
metadata:
  name: netshoot-client
  labels:
    app: netshoot-client
spec:
  nodeSelector:
    kubernetes.io/hostname: kind-worker
  containers:
    - name: netshoot
      image: nicolaka/netshoot
      command: ["sleep", "infinity"]
```

Let's deploy it and verify it is scheduled on the kind-worker node:

```
$ kubectl apply -f netshoot-client-pod.yaml
pod/netshoot-client created
```

Next, verify that the pod was scheduled and deployed:

```
$ kubectl get pods -o wide
```

The output should tell you the pod was deployed to the kind-worker node and that Cillium assigned it an IP address (10.244.1.67) from the kind-worker PodCIDR.

Verify you can connect from netshoot-client (located on kind-worker) to the nginx-server (located on kind-worker2).

```
$ kubectl exec -t pod/netshoot-client -- curl -s http://10.244.3.167
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
[OUTPUT TRUNCATED]
```

To verify HTTP connectivity, we really only need the HTTP status code. In our case, we expect a `200` to show a successful access. The following curl command will only output the HTTP status code.

```
$ kubectl exec -t pod/netshoot-client -- \
    curl -s -o /dev/null \
    -w "%{http_code}\n" \
    http://10.244.3.167
200
```

We've successfully deployed two pods across different nodes in the cluster, confirmed IP address assignments, and verified successful inter-node connectivity, using Cil-

ium's VXLAN-based tunneling. You will dive deeper into VXLAN and the datapath in Chapter 5.

While a successful connection from a client to the IP of a destination pod is a good first step, you might already know it's not necessarily the preferred approach in Kubernetes. Kubernetes provides the Service abstraction as a deterministic method to accessing applications.

Upon creation, Kubernetes will assign a Service a virtual IP address. When a client connects to the Service's virtual IP, traffic will be forwarded to one of the Service's endpoints.

Kubernetes uses Endpoints and EndpointSlices to track the backends of a Service. These objects are used for load-balancing decisions for each Service. EndpointSlices, in particular, offer a more scalable way to track backends across large clusters. By contrast, Cilium uses CiliumEndpoints (CEPs) and CiliumEndpointSlices (CESs) to support network routing and policy enforcement.

To better understand how Kubernetes Services work, let's create one using the following manifest (*nginx-service.yaml*). This Service listens on TCP port 80 and load-balances traffic randomly towards pods labeled `app:nginx`.

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: ClusterIP
```

Let's deploy it:

```
$ kubectl apply -f nginx-service.yaml
service/nginx-service created
```

Now that you have deployed the service, let's inspect it:

```
$ kubectl get svc nginx-service -o yaml
apiVersion: v1
kind: Service
metadata:
[...]
  name: nginx-service
  namespace: default
```

```
spec:
  clusterIP: 10.96.242.74
  clusterIPs:
  - 10.96.242.74
  internalTrafficPolicy: Cluster
  ipFamilies:
  - IPv4
  ipFamilyPolicy: SingleStack
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: nginx
  sessionAffinity: None
  type: ClusterIP
```

The preceding output highlights a few key aspects:

- The Service type is ClusterIP, which means it's only accessible from within the cluster
- It was assigned a cluster-internal IP (10.96.242.74) in the 10.96.0.0/16 CIDR range (the default IPv4 Service Subnet in kind)

You can confirm that the Service is working by sending a HTTP request to its ClusterIP:

```
$ kubectl exec netshoot-client -- curl http://10.96.242.74
<h1>Welcome to nginx!</h1>
```

ClusterIP is one of several Kubernetes Services, alongside NodePort and LoadBalancer. Chapter 5 will provide a refresher on Service types. While ClusterIP is used for internal communication, most of your applications will need to be accessible from outside the cluster. Chapter 7 will introduce Ingress and Gateway API for handling external traffic, and Chapter 10 will explain the networking options available to expose your applications.

Although the client sends traffic to the Service IP, the actual response comes from one of the backend Pods created by the deployment. Kubernetes uses the EndpointSlice API to track which pods back a given Service. Each EndpointSlice lists the IPs of the pods that match the Service's label selector.

```
$ kubectl get endpointslices.discovery.k8s.io nginx-service-66q4l
NAME                  ADDRESSTYPE   PORTS   ENDPOINTS      AGE
nginx-service-66q4l   IPv4          80      10.244.1.189   2m13s
```

If you scale the Deployment, Kubernetes automatically updates the endpoints.

```
$ kubectl scale deployment nginx-deployment --replicas=2
deployment.apps/nginx-deployment scaled
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
netshoot-client                     1/1     Running   0          7m15s
nginx-deployment-979f5455f-9pm7d    1/1     Running   0          4m29s
nginx-deployment-979f5455f-xw8nz    1/1     Running   0          13s
$ kubectl get endpointslices.discovery.k8s.io nginx-service-66q4l
NAME                  ADDRESSTYPE   PORTS   ENDPOINTS                   AGE
nginx-service-66q4l   IPv4          80      10.244.1.189,10.244.1.89    14m
```

After scaling, you'll see both pods running:

```
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
netshoot-client                     1/1     Running   0          7m15s
nginx-deployment-979f5455f-9pm7d    1/1     Running   0          4m29s
nginx-deployment-979f5455f-xw8nz    1/1     Running   0          13s
$ kubectl get endpointslices.discovery.k8s.io nginx-service-66q4l
NAME                  ADDRESSTYPE   PORTS   ENDPOINTS                   AGE
nginx-service-66q4l   IPv4          80      10.244.1.189,10.244.1.89    14m
```

And the EndpointSlices object now reflects both IPs:

```
$ kubectl get endpointslices.discovery.k8s.io nginx-service-66q4l
NAME                  ADDRESSTYPE   PORTS   ENDPOINTS                   AGE
nginx-service-66q4l   IPv4          80      10.244.1.189,10.244.1.89    14m
```

Behind the scenes, these endpoint IPs are used by Kubernetes to direct traffic to Pods. By default, clusters like kind use kube-proxy, a DaemonSet that runs on every node and programs iptables (with IPVS and nftables as newer alternatives) rules to implement Service routing.

You can confirm that kube-proxy is running:

```
$ kubectl get -n kube-system daemonsets/kube-proxy
NAME         DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELEC-
TOR           AGE
kube-proxy   3         3         3       3            3           kubernetes.io/
os=linux   42m
```

In Chapter 6, you'll learn how Cilium replaces kube-proxy entirely using eBPF, eliminating the need for iptables, IPVS, or nftables altogether.

Another benefit of leveraging a Service is the automatic service DNS naming generation.

You can verify successful DNS resolution by accessing the nginx-service name from the client:

```
$ kubectl exec netshoot-client -- curl http://nginx-service.default.svc.clus-
ter.local
<h1>Welcome to nginx!</h1>
```

Given that the client and the nginx servers reside in the same namespace, you only need the service name:

```
$ kubectl exec netshoot-client -- curl http://nginx-service
<h1>Welcome to nginx!</h1>
```

Cilium is not responsible for DNS activities, however, as we saw briefly in Chapter 2, it sometimes leverages a component named DNS Proxy when users define domain-based network policies. You will learn more about it in Chapter 12.

Securing the Application with Cilium Network Policies

With the application deployed and reachable, let's apply a network policy to restrict access-allowing only trusted systems to access it.

We will create a network policy that only allows access from the `netshoot-client` to the `nginx-server` pod, blocking the traffic from unauthorized systems.

First, to help us demonstrate network policies, we'll create another client pod with the following manifest (`unauthorized-client.yaml`).

```
apiVersion: v1
kind: Pod
metadata:
  name: unauthorized-client
spec:
  containers:
    - name: netshoot
      image: nicolaka/netshoot
      command: ["sleep", "infinity"]
```

Deploy it and verify it has access to the `nginx-server` prior to deploying network policies.

```
$ kubectl apply -f unauthorized-client.yaml
pod/unauthorized-client created

$ kubectl exec unauthorized-client -- curl http://nginx-service
<h1>Welcome to nginx!</h1>
```

You might already be familiar with Kubernetes Network Policies. Cilium Network Policies (CNP) build upon them and provide more granular controls. Chapter 12 will cover the concepts of identity, endpoints, labels and how to construct CNPs. For now just know that the approach to define CNPs is through the use of Kubernetes labels.

Now, take a look at the following network policy (`policy.yaml`).

```
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
```

```
metadata:
  name: ch03-policy
  namespace: default
spec:
  endpointSelector:
    matchLabels:
      app: nginx
  ingress:
    - fromEndpoints:
        - matchLabels:
            app: netshoot-client
      toPorts:
        - ports:
            - port: "80"
```

We'll cover the anatomy of a Cilium Network Policy in detail in Chapter 12 but, let's do a quick review:

- `endpointSelector` selects the pods the policy applies to (e.g., `app:nginx`).
- `ingress` defines in which traffic direction the policy applies to. In this instance, we are defining a rule that will authorize traffic *to* the pods with the `app:nginx` label (egress would instead apply to traffic leaving the pods).
- `fromEndpoints` determines which source pods are allowed to connect (those with the `app:netshoot-client` label)
- `toPorts` specifies which TCP/UDP ports are permitted.

By default, all egress and ingress traffic is allowed for all endpoints. When an endpoint is selected by a network policy, it transitions to a default-deny state, where only explicitly allowed traffic is permitted.

Given that our rule has an ingress section, the endpoint goes into default deny-mode for ingress.

In other words, when traffic enters endpoints with the `app:nginx` labels, only traffic from clients with the `app:netshoot-client` to the port 80 will be allowed. Any other traffic to these particular endpoints will be denied.

Isovalent offers a free SaaS visualization Network Policy Editor tool on *https://editor.cilium.io* to help engineers create and manage network policies. You can even upload policies like the one above to see how they would take effect (see FIGURE 3-1).

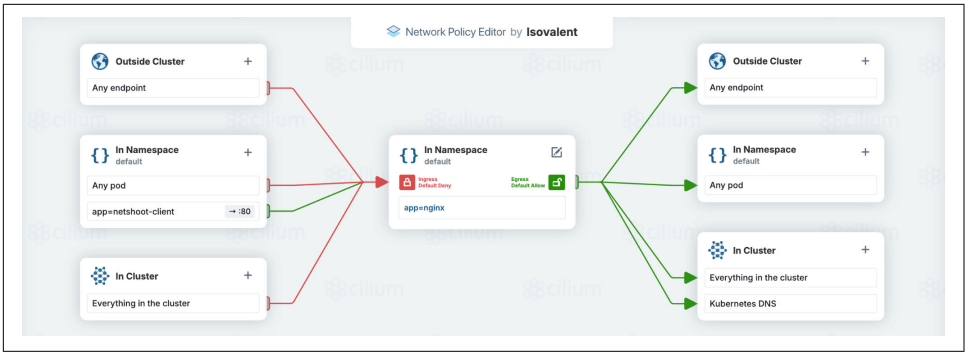

Figure 2-1. Network Policy Editor

Let's check that our `netshoot-client` has the right label and that `unauthorized-client` has not.

```
$ kubectl get pods --show-labels
NAME                     READY   STATUS    RESTARTS   AGE     LABELS
netshoot-client          1/1     Running   0          2d20h   app=netshoot-
client
nginx-server             1/1     Running   0          3d3h    app=nginx
unauthorized-client      1/1     Running   0          80m     <none>
```

You can deploy the policy with the following command.

```
$ kubectl apply -f policy.yaml
ciliumnetworkpolicy.cilium.io/policy created
```

Verify that the unauthorized-client pod can no longer access the nginx server, while the netshoot-client still can.

```
$ kubectl exec unauthorized-client -- curl --max-time 3 -s http://nginx-service
command terminated with exit code 28
$ kubectl exec netshoot-client -- curl --max-time 3 -s http://nginx-service
<h1>Welcome to nginx!</h1>
```

Unlike standard Kubernetes network policies, Cilium Network Policies can also apply at a greater depth, up to layer 7. It is particularly useful to secure API calls, such as REST API or gRPC. With layer 7 network policies, you can restrict access based on HTTP methods, headers, and paths. You can verify now by augmenting the previous network policy and limit the HTTP call to only the `/index.html` file.

Traffic matching layer 7 rules is directed to Envoy, which performs HTTP inspection before forwarding or denying the request. We introduced Envoy in Chapter 2 and you will learn more about how traffic is re-directed to Envoy in Chapter 15 (Life of a Packet).

By default, the `NGINX` server we deployed also includes an error page named 50x.html. The current Cilium Network Policy lets you access it, with no restrictions.

```
$ kubectl exec -t netshoot-client -- curl http://nginx-service/50x.html
<html>
<head>
<title>Error</title>
</head>
<body>
<h1>An error occurred.</h1>
<p>Sorry, the page you are looking for is currently unavailable.<br/>
Please try again later.</p>
<p>If you are the system administrator of this resource then you should check
the error log for details.</p>
<p><em>Faithfully yours, nginx.</em></p>
</body>
```

You can refine the previous policy by only allowing access to the *index.html* page by adding layer 7 rules such as the following:

```
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
metadata:
  name: policy
  namespace: default
spec:
  endpointSelector:
    matchLabels:
      app: nginx
  ingress:
    - fromEndpoints:
        - matchLabels:
            app: netshoot-client
      toPorts:
        - ports:
            - port: "80"
              protocol: TCP
          rules: <1>
            http: <1>
            - method: "GET" <1>
              path: "/index.html" <1>
```

<1>This ensures that, in addition to the previous rules and conditions we specified earlier, only access with the HTTP GET Method to the *index.html* page is permitted.

Deploy the updated network policy (it will overwrite the one previously configured):

```
$ kubectl apply -f policy-with-l7.yaml
ciliumnetworkpolicy.cilium.io/policy configured
```

Verify access to /50x.html and */index.html*. The former is now blocked while the latter is still allowed.

```
$ kubectl exec netshoot-client -- curl -s http://nginx-service/50x.html
Access denied
$ kubectl exec netshoot-client -- curl http://nginx-service/index.html
<title>Welcome to nginx!</title>
```

You might have noticed that this time, you received an "Access Denied" error, rather than a timeout as you saw earlier when you used a Layer 3/4 policy. While enforced L3/L4 rules lead to dropped packets, enforced layer 7 rules applied by Envoy return deny codes for the application protocol. In this case, we're parsing HTTP traffic so it returns `403` for a denied request.

Now that we have set up a security policy to control who can access our applications, let's observe its effect on actual flows.

Monitoring the Application with Cilium Hubble

Hubble is the observability subsystem of Cilium. You cannot use Hubble without Cilium because it leverages the data collected through eBPF to provide deep visibility into network traffic and policy enforcement.

By default, Hubble runs on the individual node on which the Cilium agent runs. This confines the network insights to the traffic observed by the local Cilium agent.

To enable Hubble to start collecting and aggregating flow logs across the entire cluster, use the following command.

```
$ helm upgrade cilium cilium/cilium --version 1.17.1 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true
```

Alternatively, you can use the Cilium CLI command `cilium hubble enable` to activate the Hubble function instead.

Once enabled, verify that the Hubble Relay status is `OK` with `cilium status`:

```
$ cilium status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:             OK
 \__/¯¯\__/    Operator:           OK
 /¯¯\__/¯¯\    Envoy DaemonSet:    OK
 \__/¯¯\__/    Hubble Relay:       OK
    \__/       ClusterMesh:        disabled

DaemonSet              cilium             Desired: 3, Ready: 3/3, Available: 3/3
DaemonSet              cilium-envoy       Desired: 3, Ready: 3/3, Available: 3/3
Deployment             cilium-operator    Desired: 1, Ready: 1/1, Available: 1/1
Deployment             hubble-relay       Desired: 1, Ready: 1/1, Available: 1/1
Containers:            cilium             Running: 3
                       cilium-envoy       Running: 3
                       cilium-operator    Running: 1
```

```
                    hubble-relay       Running: 1
[...]
```

To access the logs collected from Hubble, you can use the port-forward feature of Kubernetes. With the following command, you can connect the Hubble client to the local port 4245 and access the Hubble Relay service in your Kubernetes cluster. The `&` lets you run Hubble in the background rather than having to keep the terminal open.

```
$ cilium hubble port-forward&
[1] 71325
  Hubble Relay is available at 127.0.0.1:4245
$
```

The equivalent kubectl command would be `$ kubectl -n kube-system port-forward svc/hubble-relay 4245:80 &`

```
$ kubectl -n kube-system port-forward svc/hubble-relay
4245:80 &
```

You can now connect to the Hubble Server, using the Hubble UI or CLI. To start with, let's use the Hubble CLI.

First, verify the status of Hubble using `hubble status`. It provides a real-time snapshot of Hubble's health, including metrics such as the rate of flows observed per second and the volume of flows currently stored locally in its ring buffer.

```
$ hubble status
Healthcheck (via localhost:4245): Ok
Current/Max Flows: 16,380/16,380 (100.00%)
Flows/s: 9.43
Connected Nodes: 3/3
```

And list all the nodes that Hubble Relay is connected to, with the following command:

```
$ hubble list nodes
NAME                           STATUS      AGE         FLOWS/S   CURRENT/MAX-
FLOWS
kind-kind/kind-control-plane   Connected   29h35m53s   1.05      4095/4095
(100.00%)
kind-kind/kind-worker          Connected   29h35m53s   1.06      4095/4095
(100.00%)
kind-kind/kind-worker2         Connected   29h35m53s   4.32      4095/4095
(100.00%)
```

Next, use `hubble observe` to list all the flows collected by Hubble.

```
$ hubble observe
Mar 14 11:22:41.654: default/netshoot-client:50976 (ID:18901) -> kube-system/
coredns-668d6bf9bc-qdkn6:53 (ID:23236) to-overlay FORWARDED (UDP)
Mar 14 11:22:41.655: default/netshoot-client:50976 (ID:18901) <- kube-system/
coredns-668d6bf9bc-qdkn6:53 (ID:23236) to-endpoint FORWARDED (UDP)
```

```
Mar 14 11:22:41.656: default/netshoot-client:36998 (ID:18901) -> default/nginx-
deployment-979f5455f-xw8nz:80 (ID:4437) policy-verdict:L3-L4 INGRESS ALLOWED
(TCP Flags: SYN)
Mar 14 11:22:41.657: default/netshoot-client:36998 (ID:18901) -> default/nginx-
deployment-979f5455f-xw8nz:80 (ID:4437) http-request FORWARDED (HTTP/1.1 GET
http://nginx-service/index.html)
Mar 14 11:22:41.658: default/netshoot-client:36998 (ID:18901) <- default/nginx-
deployment-979f5455f-xw8nz:80 (ID:4437) http-response FORWARDED (HTTP/1.1 200
1ms (GET http://nginx-service/index.html))
```

Like `tcpdump`, the Hubble CLI can be extremely verbose!

The output shows traffic from the netshoot client pod, including a DNS query to CoreDNS (UDP/53) in a different namespace, followed by a successful HTTP request to the NGINX service:

You can see, with the arrow direction (`<-` and `->`), the flow of the traffic, including which namespace (`default` and `kube-system`) it originated from and where it was sent. The logs also show the identity of each pod and whether traffic was forwarded, dropped or denied by a network policy.

The first two lines show a DNS query to CoreDNS. Notice the `to-overlay` flag indicating the path the packet has followed: as the DNS server was located on a different node than the client, the traffic was sent via the VXLAN tunnel.

Incidentally, sending DNS traffic outside of the node is inefficient as it adds latency and creates unnecessary inter-node traffic. In Chapter 8 you will learn how Cilium can optimize DNS by intercepting these requests locally using a feature called Local Redirect Policy.

To control the output of Hubble, let's use filters to narrow down the output. Let's only list the flows originating from netshoot-client.

```
$ hubble observe --from-pod netshoot-client
Jul  2 10:39:11.325: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-overlay FORWARDED (TCP Flags: SYN)
Jul  2 10:39:11.325: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) policy-verdict:L3-L4 INGRESS ALLOWED
(TCP Flags: SYN)
Jul  2 10:39:11.325: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-proxy FORWARDED (TCP Flags: SYN)
Jul  2 10:39:11.325: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-overlay FORWARDED (TCP Flags: ACK)
Jul  2 10:39:11.325: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-proxy FORWARDED (TCP Flags: ACK)
Jul  2 10:39:11.326: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-overlay FORWARDED (TCP Flags: ACK,
PSH)
```

```
Jul  2 10:39:11.326: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-proxy FORWARDED (TCP Flags: ACK,
PSH)
Jul  2 10:39:11.326: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) http-request FORWARDED (HTTP/1.1 GET
http://nginx-service/index.html)
Jul  2 10:39:11.327: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-overlay FORWARDED (TCP Flags: ACK,
FIN)
Jul  2 10:39:11.327: default/netshoot-client:42214 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) to-proxy FORWARDED (TCP Flags: ACK,
FIN)
```

The output was taken shortly after executing the `kubectl exec netshoot-client -- curl nginx-service/index.html` command. The `to-proxy` confirms that traffic was redirected to the Envoy proxy because of the layer 7 policy in place.

You can combine multiple filters together, for example, to only show traffic from netshoot-client to a HTTP path "/index.html":

```
$ hubble observe --from-pod netshoot-client --http-path "/index.html"
Jul  2 10:41:55.108: default/netshoot-client:59110 (ID:7613) -> default/nginx-
deployment-979f5455f-qcbjh:80 (ID:42054) http-request FORWARDED (HTTP/1.1 GET
http://nginx-service/index.html)
```

Another useful Hubble CLI filter, `--label`, lets you select flows based on labels:

```
$ hubble observe --label app=nginx
[...]
Mar 14 11:28:32.002: 10.244.1.218:51932 (ID:18901) -> default/nginx-
deployment-979f5455f-9pm7d:80 (ID:4437) to-endpoint FORWARDED (TCP Flags: ACK,
FIN)
Mar 14 11:28:32.002: 10.244.1.218:51932 (host) <- default/nginx-
deployment-979f5455f-9pm7d:80 (ID:4437) to-stack FORWARDED (TCP Flags: ACK, FIN)
```

You can also filter based on the network policy verdict and only check the flows where a network policy decision was made. You can find the full list of available filters with `hubble observe -help`.

```
$ hubble observe -t policy-verdict
Mar 21 11:18:51.917: default/netshoot-client:34898 (ID:11661) -> default/nginx-
deployment-979f5455f-bnxh7:80 (ID:5834) policy-verdict:L3-L4 INGRESS ALLOWED
(TCP Flags: SYN)
Mar 21 12:12:27.525: default/netshoot-client-worker2:48768 (ID:11661) ->
default/nginx-deployment-979f5455f-bnxh7:80 (ID:5834) policy-verdict:L3-L4
INGRESS ALLOWED (TCP Flags: SYN)
Mar 21 12:15:40.016: default/unauthorized-client:41212 (ID:8087) <> default/
nginx-deployment-979f5455f-bnxh7:80 (ID:5834) policy-verdict:none INGRESS
DENIED (TCP Flags: SYN)
```

Hubble CLI provides insight into what's happening in your cluster but for those of you who prefer a more visual representation of the network traffic, you can

also leverage Hubble UI. In Chapter 2 we mentioned how the Hubble UI can also consume the flows aggregated from the Hubble Relay.

Use the following Helm commands to enable the Hubble UI:

```
$ helm upgrade cilium cilium/cilium --version 1.17.1 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true \
   --set hubble.ui.enabled=true
```

If you prefer to use the Cilium CLI to enable the UI (via `cilium hubble enable --ui`), you must first temporarily disable Hubble using `cilium hubble disable`. This is necessary because the Hubble UI cannot be enabled once Hubble is already running.

Once you enable Hubble UI, you will see a Hubble UI Deployment being created and a Hubble UI pod made of a backend container and a frontend UI.

```
$ kubectl -n kube-system get deployment/hubble-ui
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
hubble-ui   1/1     1            1           2m22s

$ kubectl get pod hubble-ui-76d4965bb6-m8s78 -n kube-system -o yaml
 containerStatuses:
  - containerID:
[...]
    name: backend
  - containerID:
[...]
    name: frontend
```

To access the Hubble UI, we will once again use the Kubernetes port-forward functionality. Note that the Hubble UI can also be accessed over Ingress, as you will learn in Chapter 14.

```
$ cilium hubble ui
   Opening "http://localhost:12000" in your browser...
```

A browser should automatically open. If it doesn't, simply open *http://localhost:12000* in your browser. You will see the Hubble UI landing page (Figure 3-2):

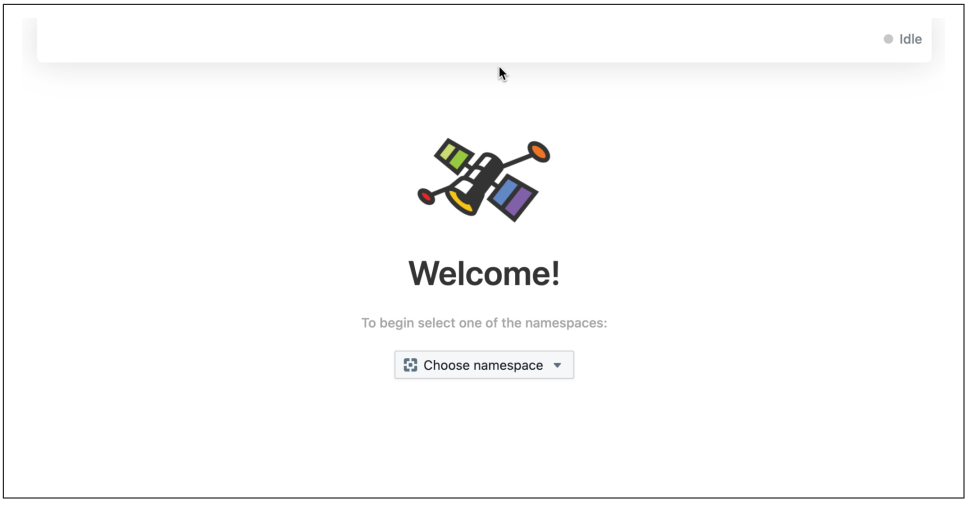

Figure 2-2. Hubble UI Landing Page

Just like the Hubble CLI, the Hubble UI can display flows filtered by namespace. Select the default namespace, and you'll see the same flows you observed earlier - now shown as a visual service map that illustrates communication between services.

Click on any of the pods to see the equivalent representation to `hubble observe -pod`. For example, you can see the flows to and from `netshoot-client` in Figure 3-3.

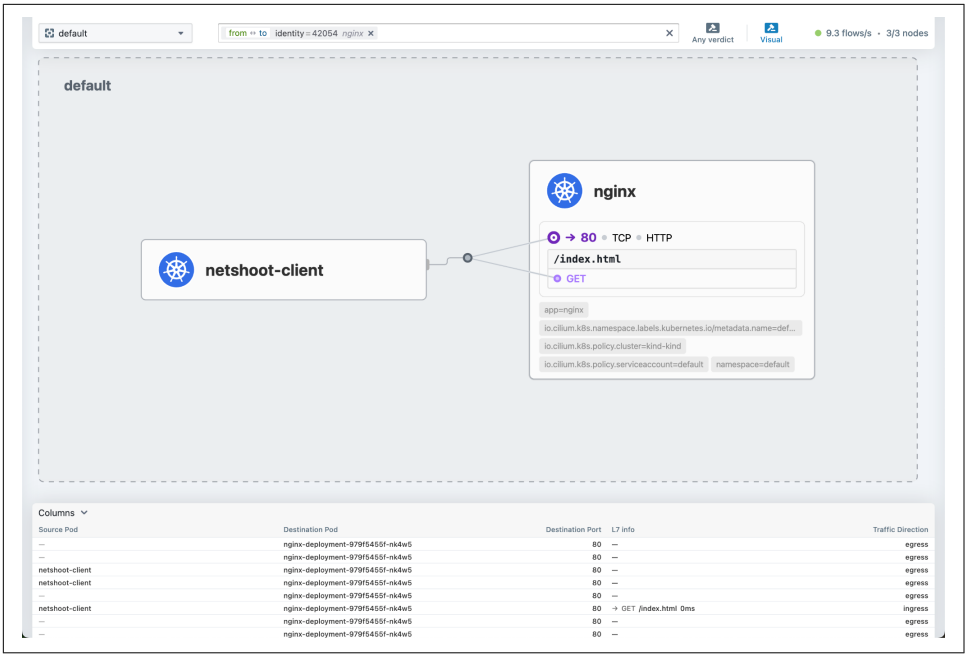

Figure 2-3. Hubble UI Service Map

You might also notice an `L7 Info` column in the user interface, populated with details such as HTTP path, method and latency. This is a side effect - and added benefit - of the Layer 7-based Cilium Network Policy we enforced earlier. Layer 7 rules not only provide more granular control - they also enhance observability. As traffic is intercepted by Envoy, it enriches the network flow information with application-layer context.

Chapter 14 will dive deeper into Hubble.

Conclusion

We hope this chapter has demonstrated that getting started with Cilium is straightforward. You've deployed Cilium in a Kubernetes cluster and taken your first steps toward using it to secure, observe, and manage application traffic.

In the next chapter, we'll take a closer look at how Cilium handles IP address management and establishes network connectivity for pods. You'll learn how to choose the right IPAM mode for your environment and understand the implications of using IPv4, IPv6, or both.

CHAPTER 3
IP Address Management

> **A Note for Early Release Readers**
>
> With Early Release ebooks, you get books in their earliest form—the author's raw and unedited content as they write—so you can take advantage of these technologies long before the official release of these titles.
>
> This will be the 4th chapter of the final book. Please note that the GitHub repo is available at *https://github.com/isovalent/cilium-up-and-running*.
>
> If you'd like to be actively involved in reviewing and commenting on this draft, please reach out to the editor at *gobrien@oreilly.com*.

Before deploying Cilium in a Kubernetes cluster, platform engineers need to make two fundamental networking decisions: how pod IP addresses are allocated, and whether to use IPv4, IPv6, or both. Selecting the right IP Address Management (IPAM) mode is essential, as changing it later may disrupt connectivity for running workloads or may not be supported at all.

Broader organizational requirements typically drive the choice between IPv4, IPv6 or dual-stack, but it also demands careful consideration by platform teams during initial cluster design.

This chapter will help you make an informed decision. We will start by examining the different IPAM modes supported by Cilium, highlighting the pros and cons of each one. We will then explore the dual-stack and IPv6 deployment models. Every aspect will come with sample manifests for you to try in your own test environments.

IPv4 and IPv6 Support in Cilium

Before we explain how IP addresses are assigned and how pods communicate with each other, we should start with a preamble on IPv6 support.

Kubernetes supports IPv4 single-stack, IPv6 single-stack, and dual-stack. However, this support depends on your chosen CNI to implement the required functionality.

IPv4/IPv6 single-stack networking is when each pod and service can be assigned either a single IPv4 or IPv6 address. With IPv4/IPv6 dual-stack networking, each pod and service can be assigned both an IPv4 and an IPv6 address. IPv4/IPv6 dual-stack on your Kubernetes cluster provides the following features:

- Dual-stack Pod networking (a single IPv4 and IPv6 address assignment per Pod)
- IPv4 and IPv6 enabled Services.
- Pod off-cluster egress routing (e.g., the Internet) via IPv4 and IPv6 interfaces.

When Cilium was first introduced, it supported only IPv6. This was a deliberate and forward-looking decision by the development team, who saw IPv6 as a natural fit for Kubernetes. Its expansive address space aligned well with the scale and dynamic nature of container workloads, where traditional IPv4 ranges can become limiting.

However, it soon became clear that most users - and the broader cloud-native ecosystem - were not yet prepared for IPv6-only clusters. Many tools, services, and operational practices were still built around IPv4. In response, the Cilium team added IPv4 support early in the project's lifecycle, making it possible to run clusters in dual-stack or IPv4-only mode.

IPv6 has remained a core part of Cilium's design. In some cases, new features were even implemented with IPv6 support before IPv4. While most of the examples in this book use IPv4 (reflecting its continued prevalence in Kubernetes environments), we include IPv6 examples where relevant and note any important caveats or considerations along the way.

IP Address Management (IPAM)

Assigning an IP address to entities is one of Cilium's core responsibilities. In non-containerized environments, you would have relied on Dynamic Host Configuration Protocol (DHCP) or SLAAC to dynamically assign IP addresses to machines. In Kubernetes, we refer to the process of assigning IP addresses to pods as IP Address Management (IPAM).

The CNI plugin is often responsible for assigning an IP address to your pod, and Cilium offers several options, which we will discuss in this section.

Pod IP Address Allocation

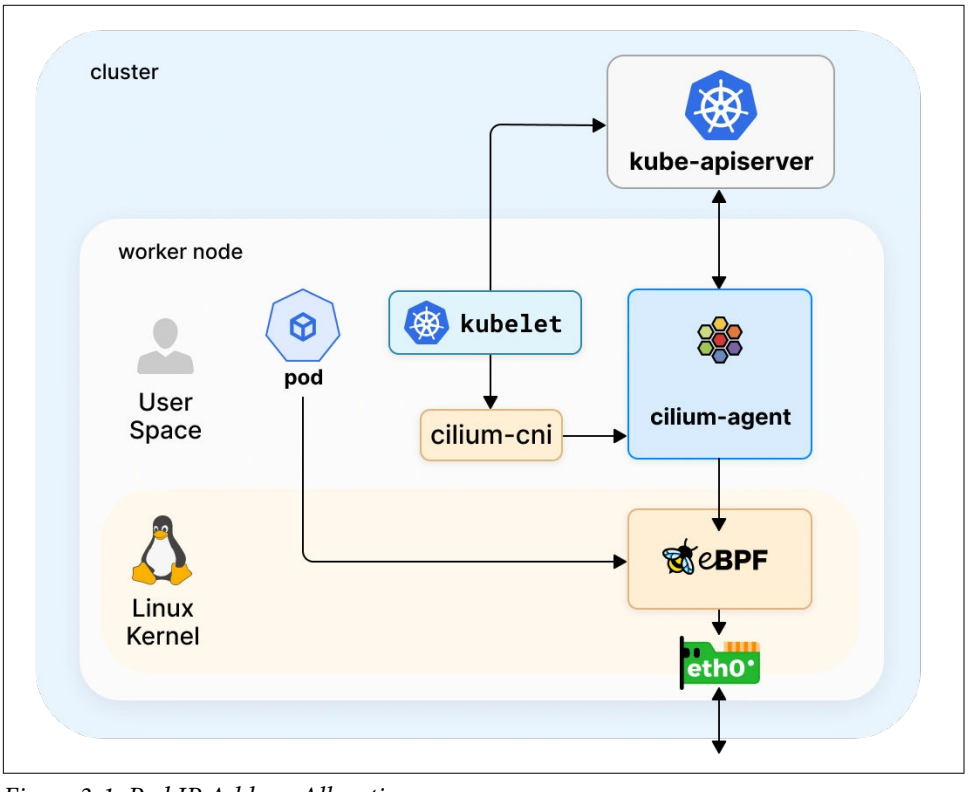

Figure 3-1. Pod IP Address Allocation

Figure 4-1 illustrates the pod IP address allocation process. When a new pod is added in Kubernetes, the Kubernetes Scheduler assigns it to a node. The Kubelet running on that node will be notified and begin creating the pod's containers.

The Kubelet uses the CNI for networking. First, the Kubelet checks the CNI configuration on the node located in */etc/cni/net.d/*. When Cilium is installed on the cluster, it creates a configuration file in */etc/cni/net.d/05-cilium.conf* which instructs the Kubelet on how to configure pod networking.

Each pod receives an IP address from a subnet referred to as PodCIDR. CIDR (classless inter-domain routing) defines IP address blocks using subnet prefixes. In Kubernetes, each node is assigned one or more PodCIDRs, and pods on that node receive their IPs from those ranges via the IPAM process.

Cilium supports multiple IPAM modes. Some are specific to a cloud provider (such as AWS and GKE), while others can be used across any Kubernetes environment.

We won't cover them all in this book, but we will cover the ones we see the most commonly used:

Kubernetes Host Scope
: Uses PodCIDRs assigned to each node by Kubernetes. Simple but inflexible.

Cluster Scope
: Cilium manages PodCIDR allocation via the operator. Default mode.

Multi-Pool
: Allows pods to be assigned IPs from multiple IP pools based on namespaces or annotations. Ideal for multi-tenant setups.

ENI IPAM
: Exclusive to AWS EKS environments. Allocates pod IPs from AWS Elastic Network Interfaces.

This chapter focuses on pod IP address management. Other components of Kubernetes require an IP address. Nodes typically receive theirs via DHCP or SLAAC. Kubernetes services receive theirs either from the Kubernetes API server or, in the case of LoadBalancer services, via a cloud provider or a dedicated system. We'll explore service networking in more detail in Chapter 6.

Kubernetes-host Scope IPAM Mode

We'll begin with Kubernetes Host Scope, the simplest IPAM mode to understand and deploy. In this mode, Kubernetes allocates a cluster-wide prefix and assigns each node a subnet from it. Cilium then assigns pod IPs from these subnets, relying on the PodCIDRs set by the Kubernetes Controller Manager.

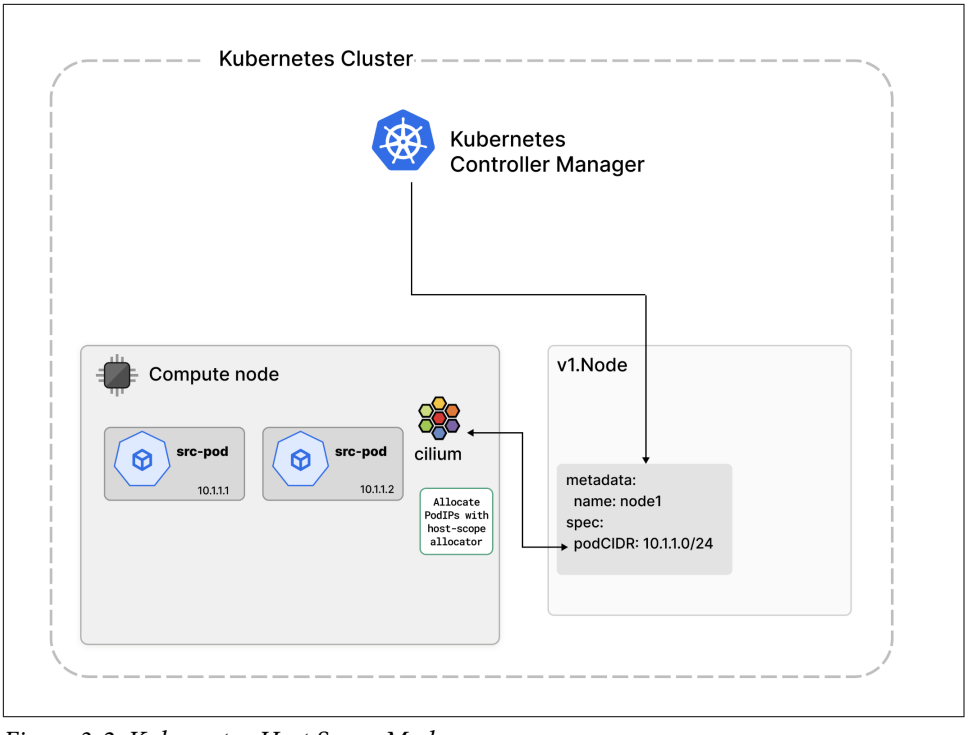

Figure 3-2. Kubernetes-Host Scope Mode

Let's try it. Start by creating a kind cluster. We will use the same kind configuration we used in Chapter 2.

You can find all the YAML manifests you will use in this chapter in the *chapter04* directory of the book's GitHub repository.

```
$ kind create cluster --config=chapter04/kind-cluster-config.yaml
```

Once you have created the cluster, install Cilium with the following Helm values (cilium-ipam-kubernetes.yaml).

```
cluster:
  name: kind-kind
ipam:
  mode: kubernetes
operator:
  replicas: 1
routingMode: tunnel
tunnelProtocol: vxlan
```

Let's use Helm to deploy Cilium.

```
$ helm install cilium cilium/cilium -n kube-system --values cilium-ipam-
kubernetes.yaml
```

By default, kind uses "10.244.0.0/16" as the cluster-wide prefix. Kubernetes assigns /24 subnets to the various nodes in the cluster. Verify in your own cluster:

```
$ kubectl get ciliumnode kind-worker -o jsonpath='{.spec.ipam}'
{"podCIDRs":["10.244.1.0/24"]}
```

The assigned prefixes may differ in your environment as subnet allocation is done randomly.

Let's now deploy a pod.

```
$ kubectl apply -f netshoot-client-pod.yaml
pod/netshoot-client created
```

You can verify with `kubectl` that the pod receives an IP address from the node's subnet.

The `kubectl get ciliumnode kind-worker -o jsonpath='{.spec.ipam}'` command indicates the PodCIDR assigned to the node and `kubectl get pod netshoot-client -o wide` displays the pod's IP address.

While Host Scope is straightforward, it comes with limitations:

- You cannot configure the size of the PodCIDRs allocated to each node.
- You cannot add or remove CIDRs to the cluster or individual nodes.

This makes it difficult to grow the pool of IP addresses available to your cluster as it grows.

Let's take a closer look at a more flexible IPAM mode.

Cluster Scope IPAM Mode

In Cluster Scope mode, Cilium (and not Kubernetes) allocates PodCIDRs to nodes. As this mode doesn't require a specific configuration of the Kubernetes cluster, it is the default IPAM option in Cilium.

In Chapter 2 we explained that one of the Cilium Operator's responsibilities is IPAM. In Cluster Scope mode, the operator assigns subnets to each node, by updating the CiliumNode custom resources (as illustrated in Figure 4-3).

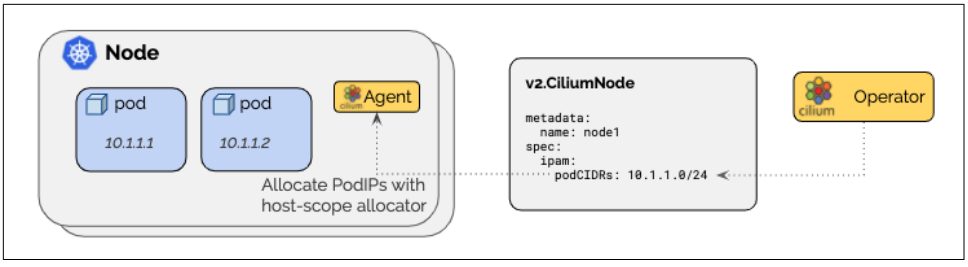

Figure 3-3. Cluster Scope IPAM Mode

First, let's re-create a cluster. If you had deployed a cluster previously, delete it first.

We recommend you delete the cluster because changing the IPAM mode in a live environment may cause persistent connectivity disruption for existing workloads.

```
$ kind delete clusters kind
$ kind create cluster --config=kind-cluster-config.yaml
```

Install Cilium with the following Helm values (cilium-ipam-cluster-scope.yaml).

```
cluster:
  name: kind-kind
ipam:
  mode: cluster-pool
operator:
  replicas: 1
routingMode: tunnel
tunnelProtocol: vxlan
```

"Cluster Scope" is sometimes referred to as "Cluster-Pool" as you can see from the Helm value.

```
$ helm install cilium cilium/cilium -n kube-system --values cilium-ipam-cluster-
scope.yaml
```

By default, Cilium uses the cluster-wide 10.0.0.0/8 prefix and assigns /24 subnets to CiliumNode resources. Verify in your cluster.

```
$ cilium config view | grep cluster-pool
cluster-pool-ipv4-cidr                    10.0.0.0/8
cluster-pool-ipv4-mask-size               24
ipam                                      cluster-pool
```

You can also list the CIDRs associated with each node.

```
$ kubectl get ciliumnode -o jsonpath='{range .items[*]}{.metadata.name}
{.spec.ipam.podCIDRs[]}{"\n"}{end}' | column -t
kind-control-plane  10.0.0.0/24
kind-worker         10.0.1.0/24
kind-worker2        10.0.2.0/24
```

Deploy a sample pod.

```
$ kubectl apply -f netshoot-client-pod.yaml
pod/netshoot-client created
```

Check the verbose output of the `cilium-dbg status` on the Cilium agent on the node hosting the pod.

```
$ kubectl -n kube-system exec -ti cilium-jhhpp -- cilium-dbg status --verbose
[...]
Allocated addresses:
  10.0.1.235 (default/netshoot-client)
  10.0.1.133 (router)
  10.0.1.238 (health)
```

One advantage cluster scope IPAM has over Kubernetes IPAM is that it can allocate multiple CIDRs to the cluster, instead of a single one. This mode also lets you specify the subnet mask size, unlike in Kubernetes-host scope mode. Cluster scope, therefore, can provide more flexibility, albeit it doesn't necessarily overcome IP address exhaustions.

Let's simulate such a scenario.

Once again, starting with a new cluster is preferable, so we will delete the previously created one and start with a new one.

Use the following values for Cilium's starting configuration (cilium-ipam-cluster-scope-small.yaml). In this example, Cilium will assign two small CIDRs (/28) to the cluster and each node will receive a /29 subnet from one of the CIDRs. As two IPs are reserved for Cilium Host and Cilium Health interfaces, each node will get 6 usage addresses.

```
cluster:
  name: kind-kind
ipam:
  mode: cluster-pool
  operator:
    clusterPoolIPv4MaskSize: 29
    clusterPoolIPv4PodCIDRList:
      - 10.0.42.0/28
      - 10.0.84.0/28
operator:
  replicas: 1
```

```
routingMode: tunnel
tunnelProtocol: vxlan
```

Let's install Cilium in this mode and observe what happens.

```
$ helm install cilium cilium/cilium -n kube-system --values cilium-ipam-cluster-
scope-small.yaml
```

Check the subnets assigned to your nodes.

```
$ kubectl get ciliumnode -o jsonpath='{range .items[*]}{.metadata.name}
{.spec.ipam.podCIDRs[*]}{"\n"}{end}' | column -t
kind-control-plane  10.0.42.8/29
kind-worker         10.0.42.0/29
kind-worker2        10.0.84.8/29
kind-worker3        10.0.84.0/29
```

As a /28 prefix only allows two /29 allocations, the third and fourth nodes receive their subnets from the second block.

Let's now create a deployment with 10 pods.

```
$ kubectl apply -f nginx-deployment-10-replicas.yaml
deployment.apps/nginx-deployment created
```

When you run `kubectl get pods -o wide`, you'll notice that a number of pods remain stuck in *ContainerCreating* stage — meaning they are not running properly. When you run `cilium-dbg status` on any of the Cilium agents, you'll see that all available IP addresses have been allocated.

```
$ kubectl exec -it ds/cilium -n kube-system -c cilium-agent -- cilium-dbg status
IPAM:                   IPv4: 6/6 allocated from 10.0.42.0/29,
```

While this mode provides greater flexibility than the Kubernetes Host Scope mode it comes with some limitations:

- You cannot dynamically add PodCIDRs to a running cluster
- You have limited control over how pods receive IP addresses.

To address these limitations, let's take a look at Multi-Pool IPAM in the next section.

Multi-Pool IPAM Mode

Multi-Pool is the most flexible IPAM mode in Cilium. It enables you to allocate PodCIDRs from multiple distinct pools, depending on user-defined workload properties, such as annotations or namespaces. Pods on the same node can receive IP addresses from different ranges, and new CIDRs can be dynamically added to a node as and when needed.

This flexibility makes Multi-Pool the recommended mode, especially for multi-tenant clusters or when granular IP addressing control is preferred. The IP allocation process used in Multi-Pool IPAM mode is described in Figure 4-4.

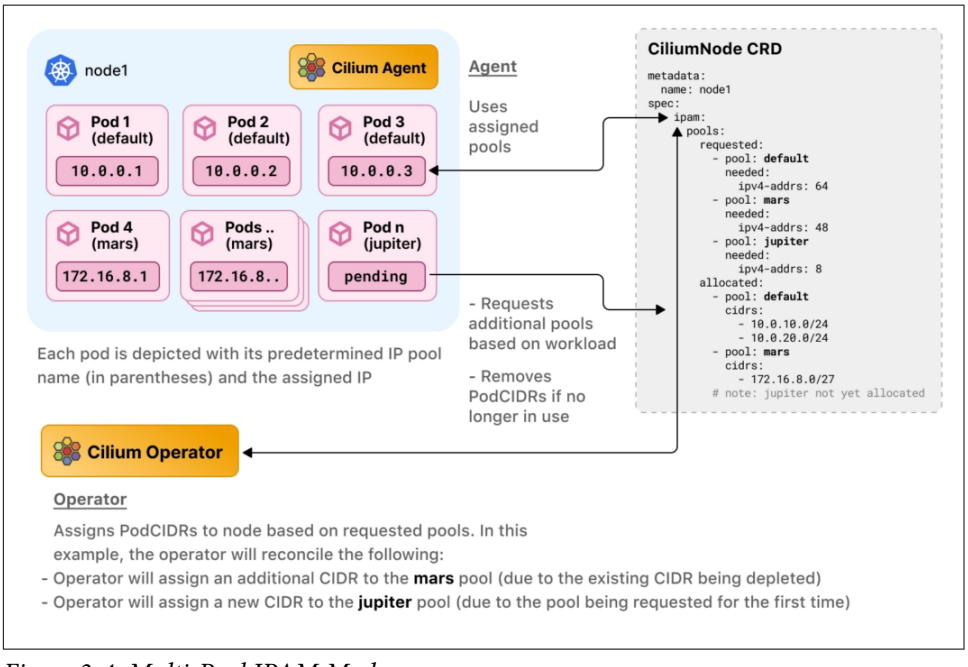

Figure 3-4. Multi-Pool IPAM Mode

Let's deploy Cilium in Multi-Pool mode. First, create a new cluster (delete the existing one if needed). Use the values below for Cilium's starting configuration (`cilium-ipam-multi-pool.yaml`).

```
ipam:
  mode: multi-pool
  operator:
    autoCreateCiliumPodIPPools:
      default:
        ipv4:
          cidrs: ["10.10.0.0/16"]
          maskSize: 27

routingMode: native
endpointRoutes:
  enabled: true

autoDirectNodeRoutes: true
ipv4NativeRoutingCIDR: 10.0.0.0/8
```

In Chapter 2, we introduced the encapsulation routing mode (using VXLAN). Multi-pool is not compatible with this mode so instead, you will need to use native routing which we will explain in more detail in Chapter 5.

Let's install Cilium.

```
$ helm install cilium cilium/cilium -n kube-system --values cilium-ipam-multi-
pool.yaml
```

The `autoCreateCiliumPodIPPools` option configures Cilium to create a default pool of IP addresses at startup and to assign nodes prefixes based on the `ipam.operator.autoCreateCiliumPodIPPools.ipv4.maskSize` value (in our case, 27).

Verify that a default pool has been created.

```
$ kubectl get ciliumpodippools.cilium.io default -o yaml
apiVersion: cilium.io/v2alpha1
kind: CiliumPodIPPool
spec:
  ipv4:
    cidrs:
    - 10.10.0.0/16
    maskSize: 27
```

Let's also verify that the kind-worker node received a /27 subnet from this pool.

```
$ kubectl get ciliumnode kind-worker -o yaml | yq .spec.ipam
pools:
  allocated:
    - cidrs:
        - 10.10.0.64/27
      pool: default
  requested:
    - needed:
        ipv4-addrs: 16
      pool: default
```

Deploy a pod on this specific node using the `netshoot-client-pod.yaml` manifest we used previously.

```
$ kubectl apply -f netshoot-client-pod.yaml
pod/netshoot-client created
```

Once more, when you verify its IP address, the pod should receive an IP address from the allocated subnet (e.g., 10.10.0.93).

```
$ kubectl get pods -o wide
NAME              READY   STATUS    RESTARTS   AGE   IP           NODE
netshoot-client   1/1     Running   0          24s   10.10.0.93   kind-worker
```

So far, Multi-Pool functions similarly to Cluster Scope IPAM. One advantage of Multi-Pool over other IPAM modes is that it gives you more control over how IP

addresses are assigned to pods. You can force a pod – or all pods in a particular namespace – to receive IPs from a particular range.

This is useful for multi-tenant environments. For example, consider two tenants: *ACME Corp.* and *Foobar Inc*. Namespaces are often used in Kubernetes to differentiate tenants.

While previous IPAM modes are not namespace-aware, Multi-Pool IPAM lets you assign pods in a namespace IPs from a distinct pool. This capability is particularly useful when traffic exits the cluster, as it helps engineers understand from which tenant the traffic originated.

In Chapter 11, you will learn more about what to consider when traffic leaves the cluster, including another method to map outbound traffic to a particular IP or interface: Egress Gateway.

Let's create two namespaces for our two tenants.

```
$ kubectl apply -f namespaces.yaml
namespace/acme-corp created
namespace/foobar-inc created
```

Create two separate pools.

```
$ cat foobar-pool.yaml
apiVersion: cilium.io/v2alpha1
kind: CiliumPodIPPool
metadata:
  name: foobar-pool
spec:
  ipv4:
    cidrs:
    - 10.30.0.0/16
    maskSize: 27
$ cat acme-pool.yaml
apiVersion: cilium.io/v2alpha1
kind: CiliumPodIPPool
metadata:
  name: acme-pool
spec:
  ipv4:
    cidrs:
    - 10.20.0.0/16
    maskSize: 27
```

Apply the manifests.

```
$ kubectl apply -f foobar-pool.yaml
ciliumpodippool.cilium.io/foobar-pool created
```

```
$ kubectl apply -f acme-pool.yaml
ciliumpodippool.cilium.io/acme-pool created
```

To ensure that all workloads in a particular namespace receive an IP address from a particular IP pool, you can annotate the namespace with `ipam.cilium.io/ip-pool=namespace_name`.

```
$ kubectl annotate namespace acme-corp ipam.cilium.io/ip-pool=acme-pool
namespace/acme-corp annotated
$ kubectl annotate namespace foobar-inc ipam.cilium.io/ip-pool=foobar-pool
namespace/foobar-inc annotated
```

When deploying workloads, they will be allocated an IP address from the appropriate pool.

```
$ kubectl apply -f nginx-deployment-10-replicas.yaml -n acme-corp
deployment.apps/nginx-deployment created

$ kubectl apply -f nginx-deployment-10-replicas.yaml -n foobar-inc
deployment.apps/nginx-deployment created
```

Here is a trimmed version of the output.

```
$ kubectl get pods -o wide -n acme-corp
NAME                               READY   STATUS    IP           NODE
nginx-deployment-979f5455f-5tkws   1/1     Running   10.20.0.25   kind-worker2
nginx-deployment-979f5455f-6dw7n   1/1     Running   10.20.0.6    kind-worker2
nginx-deployment-979f5455f-j42nl   1/1     Running   10.20.0.18   kind-worker2
nginx-deployment-979f5455f-ksvd5   1/1     Running   10.20.0.20   kind-worker2
nginx-deployment-979f5455f-l4gwg   1/1     Running   10.20.0.7    kind-worker2
nginx-deployment-979f5455f-lblmh   1/1     Running   10.20.0.12   kind-worker2
nginx-deployment-979f5455f-lbltm   1/1     Running   10.20.0.26   kind-worker2
nginx-deployment-979f5455f-lnwjt   1/1     Running   10.20.0.15   kind-worker2
nginx-deployment-979f5455f-vv4nc   1/1     Running   10.20.0.27   kind-worker2
nginx-deployment-979f5455f-xg5h9   1/1     Running   10.20.0.29   kind-worker2

$ kubectl get pods -o wide -n foobar-inc
NAME                               READY   STATUS    IP           NODE
nginx-deployment-979f5455f-2p7mc   1/1     Running   10.30.0.28   kind-worker2
nginx-deployment-979f5455f-82g98   1/1     Running   10.30.0.13   kind-worker2
nginx-deployment-979f5455f-cdnz7   1/1     Running   10.30.0.29   kind-worker2
nginx-deployment-979f5455f-hxxtt   1/1     Running   10.30.0.25   kind-worker2
nginx-deployment-979f5455f-jklr2   1/1     Running   10.30.0.14   kind-worker2
nginx-deployment-979f5455f-kndrs   1/1     Running   10.30.0.27   kind-worker2
nginx-deployment-979f5455f-pbdtt   1/1     Running   10.30.0.6    kind-worker2
nginx-deployment-979f5455f-pcqsm   1/1     Running   10.30.0.20   kind-worker2
nginx-deployment-979f5455f-qmfwg   1/1     Running   10.30.0.4    kind-worker2
nginx-deployment-979f5455f-vxdx6   1/1     Running   10.30.0.30   kind-worker2
```

Check the Cilium Node to see which PodCIDRs were assigned to the nodes.

```
$ kubectl get ciliumnode kind-worker2 -o yaml | yq .spec.ipam
pools:
  allocated:
```

```
    - cidrs:
        - 10.20.0.0/27
      pool: acme-pool
    - cidrs:
        - 10.10.0.96/27
      pool: default
    - cidrs:
        - 10.30.0.0/27
      pool: foobar-pool
  requested:
    - needed:
        ipv4-addrs: 10
      pool: acme-pool
    - needed:
        ipv4-addrs: 16
      pool: default
    - needed:
        ipv4-addrs: 10
      pool: foobar-pool
```

If you need more IPs, Cilium dynamically assigns additional subnets to the node. For example, scale up the ACME deployment from 10 replicas to 40.

```
$ kubectl scale -n acme-corp deployment nginx-deployment --replicas=40
deployment.apps/nginx-deployment scaled
```

Note the `needed` field in the following output. As the number of required IPs significantly increased, another /27 subnet from the cluster-wide `10.20.0.0/16` was allocated to the node.

```
$ kubectl get ciliumnode kind-worker2 -o yaml | yq .spec.ipam
pools:
  allocated:
    - cidrs:
        - 10.20.0.0/27
        - 10.20.0.32/27
      pool: acme-pool
    - cidrs:
        - 10.10.0.96/27
      pool: default
    - cidrs:
        - 10.30.0.0/27
      pool: foobar-pool
  requested:
    - needed:
        ipv4-addrs: 40
      pool: acme-pool
    - needed:
        ipv4-addrs: 16
      pool: default
    - needed:
        ipv4-addrs: 10
      pool: foobar-pool
```

In some environments (e.g., telecoms or financial), pods may have multiple interfaces (for example, for out-of-band management and for signaling or data). The Isovalent Enterprise Platform extends Multi-Pool with a feature called Multi-Network, which lets you connect a pod to multiple networks.

CRD-Backed IPAM Mode

Cilium supports a lesser-known IPAM mode: CRD-backed IPAM, which delegates IP address management to an external operator. This mode gives you significant flexibility, but can be complex. When using CRD-backed mode, Cilium is not automatically programmed to route traffic to the assigned CIDRs, so administrators must assign IP ranges to nodes and ensure proper routing between them.

CRD-backed IPAM is often used in managed Kubernetes environments, where the cloud provider handles IP address management and routing outside of Cilium.

The final IPAM mode we will review is specific to Amazon Elastic Kubernetes Service and is widely adopted by AWS users, therefore deserving a detailed look through in this book.

ENI Mode

In ENI IPAM mode, Cilium integrates tightly with AWS networking by assigning pod IP addresses from secondary IPs on Elastic Network Interfaces (ENI). At startup, Cilium contacts the EC2 metadata ENI to retrieve instance ID, type, and VPC information and populates each Cilium Node custom resource with the info (as illustrated in Figure 4-5).

Cilium then allocates pod IPs directly from the pool of secondary addresses associated with those ENIs. This provides a clear benefit: pods receive IPs that are directly routable within the AWS VPC. No NAT is required, which simplifies networking and improves observability for operators.

Note that ENI Mode is not currently supported with IPv6.

ENI mode is only relevant when running Cilium on Amazon EKS, where ENIs and their IPs are managed by AWS.

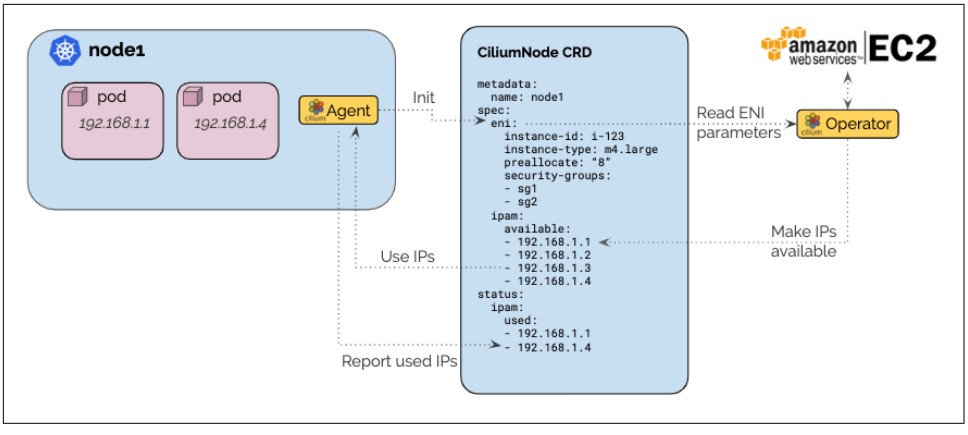

Figure 3-5. ENI Mode

Let's explain it by walking through the example.

Unlike the previous examples where we were using kind, deploying a cluster on EKS comes with a cost. Proceed carefully before deploying a managed cloud-hosted cluster.

To deploy an EKS cluster, we recommend the eksctl tool. Follow the instructions in the documentation to install and configure it.

The eksctl configuration below will create a cluster with 2 nodes, in the `eu-west-1` region (update it to the region of your choice).

```
apiVersion: eksctl.io/v1alpha5
kind: ClusterConfig

metadata:
  name: cuar
  region: eu-west-1

managedNodeGroups:
- name: ng-1
  desiredCapacity: 2
  privateNetworking: true
  taints:
   - key: "node.cilium.io/agent-not-ready"
     value: "true"
     effect: "NoExecute"
```

In certain cloud environments (like EKS or GKE), another CNI plugin might already be installed, which may cause Cilium to fail to take control of CNI. To address this, Cilium supports a taint-based mechanism that delays pod scheduling until the agent is ready, helping ensure pods are managed by Cilium rather than the existing CNI plugin. Read more about it in the Cilium documentation.

Deploy the cluster

```
$ eksctl create cluster -f ./eks-config.yaml
```

After 15-20 minutes, the cluster should be ready, and you can now install Cilium.

```
$ helm install cilium cilium/cilium -n kube-system --values helm-eni-values.yaml
```

Let's deploy five pods with a sample deployment. All pods should receive an IP address as you would expect.

```
$ kubectl apply -f echo-deployment.yaml
deployment.apps/echoserver created

$ kubectl get pods -o wide
NAME                          READY   STATUS    IP
echoserver-79974b75cd-fwd9r   1/1     Running   192.168.168.230
echoserver-79974b75cd-kn9fz   1/1     Running   192.168.156.61
echoserver-79974b75cd-p5q9b   1/1     Running   192.168.131.53
echoserver-79974b75cd-wgdzf   1/1     Running   192.168.170.2
echoserver-79974b75cd-whjvc   1/1     Running   192.168.142.65
```

The pod IPs are actually taken from the EC2 instance's network interface and listed as a secondary private IPs, as you can see on the AWS Console, under EC2 > Instances > instance_name > Networking (Figure 4-6).

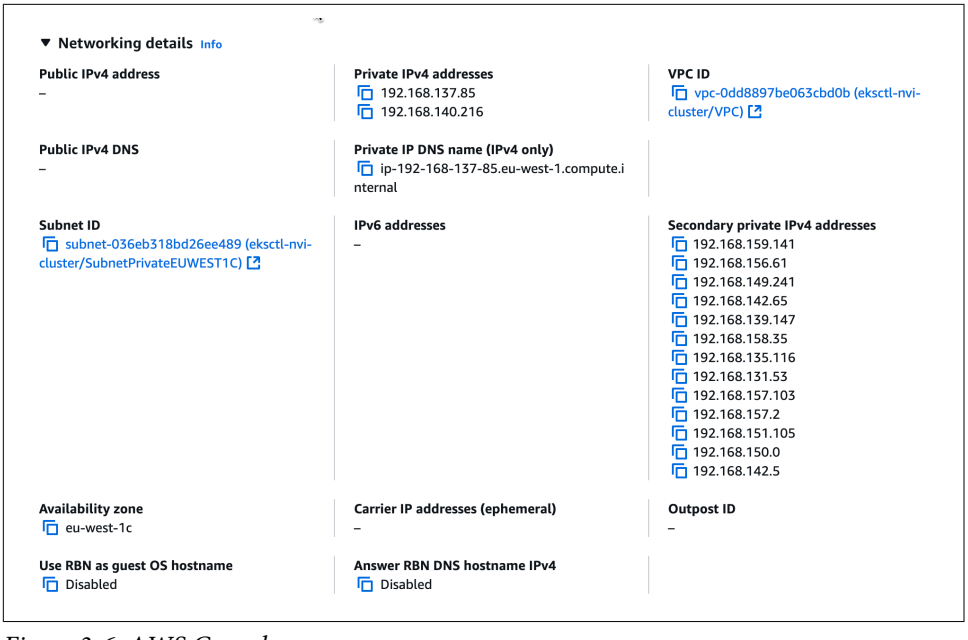

Figure 3-6. AWS Console

You can also see the IPs being populated on the Cilium Node resource.

```
$ kubectl get cn ip-192-168-132-54.eu-west-1.compute.internal -o yaml | yq .sta-
tus.eni
enis:
  eni-02f39382678db84b5:
    addresses:
      - 192.168.159.141
      - 192.168.156.61
      - 192.168.149.241
      - 192.168.142.65
      - 192.168.139.147
      - 192.168.158.35
      - 192.168.135.116
      - 192.168.131.53
      - 192.168.157.103
[...]
```

When using ENI mode, you should be aware of certain scaling limitations. In ENI mode, the number of IPs that can be allocated per node is limited by the instance's type ENI and secondary IP capacity (e.g., a m5.large supports only 3 ENIs with 10 IPs each). This restricts the number of pods a node can run (with a m5.large only able to support 30 pods - far below Kubernetes' 110 pods per node limit).

Cilium supports AWS Prefix Delegation, a feature that assigns entire CIDR blocks (/28) to each ENI. With this feature, pod density per node increases significantly, without having to scale the cluster horizontally or vertically.

IPv6 Clusters

The other essential decision you will need to make regarding IP addressing is which version of IP to use: IPv4, IPv6, or both. While this choice often stems from broader organizational directives, it has direct implications for your Kubernetes networking architecture.

Cilium offers robust IPv6 support across all IPAM modes described in this chapter, with the exception of ENI mode. The vast majority of Cilium features are IPv6-ready, although a few still have limitations: encapsulation routing mode (covered in Chapter 5) and Egress Gateway (covered in Chapter 11) currently do not support IPv6.

In this section, we will explore two common deployment models:

- *Dual Stack (IPv4 and IPv6)*, where pods and services receive both an IPv4 and IPv6 address
- *Single Stack (IPv6 only)*, where pods and services only receive an IPv6 address.

Choosing between IPv4, Dual Stack, or IPv6-only should be based on a combination of technical readiness and organizational strategy. You should consider the following aspects:

Organizational IPv6 strategy: Is your company actively working towards IPv6 adoption or still primarily reliant on IPv4? Some enterprises mandate IPv6 for all new workloads; in some regions, governments and public sector institutions go even further, requiring IPv6 compliance for all digital infrastructure and services.

Operational readiness: Do your teams have the tooling and expertise to monitor and troubleshoot IPv6 traffic? Logging, observability, and security tooling must be IPv6-aware.

Ecosystem limitations: While Kubernetes and Cilium support IPv6 well, other tools and services may not. For instance, as of writing, GitHub-hosted services are not

accessible via IPv6. This means IPv6-only environments may need to rely on mechanisms like NAT64/DNS64[1] to reach IPv4-only endpoints, adding complexity.

As a general guide:

- IPv4-only is the default and safest option for most organizations. It ensures broad compatibility albeit does not help you future-proof your stack.
- Dual Stack is often a transitional strategy. It allows IPv6 experimentation while maintaining full IPv4 compatibility. However, it introduces operational complexity, especially in IP management and observability.
- IPv6-only is viable in greenfield deployments where external dependencies are well understood and NAT64/DNS64 can be reliably deployed. It demands a mature operational model and close awareness of third-party IPv6 support.

With those considerations in mind, let's look at how to configure and operate Cilium in both Dual Stack and IPv6-only modes.

Dual Stack (IPv4/IPv6)

In Dual Stack configuration, each pod is allocated both an IPv4 and an IPv6 address. This enables communications with both IPv6 systems and legacy apps that only use IPv4.

To test it, we will use the following kind configuration (kind-cluster-dual-stack.yaml).

Use the following for Linux. Notice that the ipFamily configuration is set to `dual` to enable dual stack functionality, supporting both IPv4 and IPv6 protocols.

```
---
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
  - role: control-plane
  - role: worker
  - role: worker
  - role: worker
networking:
  disableDefaultCNI: true
  ipFamily: dual
```

1 NAT64 and DNS64 are mechanisms that allow IPv6-only clients to communicate with IPv4-only servers. DNS64 synthesizes AAAA records from A records, enabling IPv6 clients to resolve IPv4-only domain names. NAT64 then translates the IPv6 traffic to IPv4 at the network edge. Together, they provide access from IPv6-only environments to legacy IPv4 services.

This configuration works on Linux machines with IPv6 support enabled. When using a Mac, add the `apiServerAddress` set to `127.0.0.1` under `networking` to work around the lack of IPv6 forwarding in Docker. Read more on the kind docs for platform-specific guidance.

Deploy the cluster.

```
$ kind create cluster --config=kind-cluster-dual-stack.yaml
```

The nodes receive both an IPv4 and IPv6 address. Verify with the following command.

```
$ kubectl get node kind-worker -o jsonpath='{.status.addresses}' | jq
[
  {
    "address": "172.18.0.4",
    "type": "InternalIP"
  },
  {
    "address": "fc00:f853:ccd:e793::4",
    "type": "InternalIP"
  },
[...]
]
```

With Cilium, IPv6 is disabled by default and has to be explicitly specified in the Helm values. Use the following command to deploy Cilium in dual stack mode.

```
$ helm install cilium cilium/cilium \
  --version 1.17.1 \
  --namespace kube-system \
  --set kubeProxyReplacement=true \
  --set k8sServiceHost=kind-control-plane \
  --set k8sServicePort=6443 \
  --set ipv6.enabled=true \
  --set ipam.mode=kubernetes
```

Run the following command to see the PodCIDRs from which IPv4 and IPv6 addresses will be allocated to your pods.

```
$ kubectl describe nodes | grep PodCIDRs
PodCIDRs:                     10.244.0.0/24,fd00:10:244::/64
PodCIDRs:                     10.244.1.0/24,fd00:10:244:1::/64
PodCIDRs:                     10.244.3.0/24,fd00:10:244:3::/64
PodCIDRs:                     10.244.2.0/24,fd00:10:244:2::/64
```

Once again, let's deploy a couple of sample pods and pin them to specific nodes to validate inter-node connectivity.

```
$ kubectl apply -f pod1.yaml -f pod2.yaml
pod/pod-worker created
pod/pod-worker2 created
```

```
$ kubectl get pods -o wide
NAME          READY   AGE     IP
pod-worker    1/1     2m31s   10.244.1.117
pod-worker2   1/1     2m31s   10.244.3.29
```

As we're using the `kubernetes` IPAM mode, IP addresses are assigned from the PodCIDR assigned to each node.

```
$ kubectl describe pod pod-worker | grep -A 2 IPs
IPs:
  IP:  10.244.1.117
  IP:  fd00:10:244:1::8204
$ kubectl describe pod pod-worker2 | grep -A 2 IPs
IPs:
  IP:  10.244.3.29
  IP:  fd00:10:244:3::833f
```

Let's verify that pods can communicate. We'll test a ping from pod-worker to pod-worker's IPv6 address. Because the pods were pinned to different nodes, it should show successful IPv6 connectivity between pods on different nodes.

```
$ IPv6=$(kubectl get pod pod-worker2 -o jsonpath='{.status.podIPs[1].ip}')
$ echo $IPv6
fd00:10:244:3::833f
$ kubectl exec -it pod-worker -- ping6 -c 5 $IPv6
PING fd00:10:244:3::833f (fd00:10:244:3::833f) 56 data bytes
64 bytes from fd00:10:244:3::833f: icmp_seq=1 ttl=63 time=0.330 ms
64 bytes from fd00:10:244:3::833f: icmp_seq=2 ttl=63 time=0.156 ms
64 bytes from fd00:10:244:3::833f: icmp_seq=3 ttl=63 time=0.133 ms
64 bytes from fd00:10:244:3::833f: icmp_seq=4 ttl=63 time=0.140 ms
64 bytes from fd00:10:244:3::833f: icmp_seq=5 ttl=63 time=0.148 ms

--- fd00:10:244:3::833f ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4087ms
rtt min/avg/max/mdev = 0.133/0.181/0.330/0.074 ms
```

Let's now deploy a Service. Note that we specify PreferDualStack in the Service configuration.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: echoserver
spec:
  replicas: 5
  selector:
    matchLabels:
      app: echoserver
  template:
    metadata:
      labels:
        app: echoserver
```

```
    spec:
      containers:
        - image: ealen/echo-server:latest
          imagePullPolicy: IfNotPresent
          name: echoserver
          ports:
            - containerPort: 80
          env:
            - name: PORT
              value: "80"
---
apiVersion: v1
kind: Service
metadata:
  name: echoserver
spec:
  ipFamilyPolicy: PreferDualStack
  ipFamilies:
    - IPv6
    - IPv4
  ports:
    - port: 80
      targetPort: 80
      protocol: TCP
  type: ClusterIP
  selector:
    app: echoserver
```

Deploy the manifest.

```
$ kubectl apply -f echo-kube-dual-stack.yaml
deployment.apps/echoserver created
service/echoserver created
```

Verify that both IPv4 and IPv6 addresses are assigned.

```
# kubectl describe svc echoserver
Name:                     echoserver
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=echoserver
Type:                     ClusterIP
IP Family Policy:         PreferDualStack
IP Families:              IPv6,IPv4
IP:                       fd00:10:96::d4d5
IPs:                      fd00:10:96::d4d5,10.96.181.159
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
Endpoints:                10.244.1.206:80,10.244.3.235:80,10.244.2.55:80 + 7
more...
[...]
```

Let's verify access to the service from the pod-worker client:

```
$ ServiceIPv6=$(kubectl get svc echoserver -o jsonpath='{.spec.clusterIP}')
$ echo $ServiceIPv6
fd00:10:96::d4d5
$ kubectl exec -i -t pod-worker -- curl -6 http://[$ServiceIPv6]/ | jq
{
  "host": {
    "hostname": "[fd00:10:96::d4d5]",
    "ip": "fd00:10:244:1::8204",
    "ips": []
  },
  "http": {
    "method": "GET",
    "baseUrl": "",
    "originalUrl": "/",
    "protocol": "http"
  },
```

While we're at it, and even if it's not Cilium's role, you can verify that the Kubernetes cluster DNS addon created an AAAA record for this service.

```
$ kubectl exec -i -t pod-worker -- nslookup -q=AAAA echoserver.default
Server:         10.96.0.10
Address:        10.96.0.10#53

Name:   echoserver.default.svc.cluster.local
Address: fd00:10:96::d4d5
```

Single Stack (IPv6 only)

Let's now explore an IPv6-only cluster with Cilium. At the time of writing, IPv6 does not support tunnel (encapsulation) mode, and therefore IPv6 can only be installed in native mode (you'll learn more about native mode in Chapter 5).

Let's start once again with the kind cluster configuration. When using IPv6, set ipfamily to `ipv6` and specify the pod and service subnets.

```
---
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
  - role: control-plane
  - role: worker
  - role: worker
networking:
  ipFamily: ipv6
  disableDefaultCNI: true
  podSubnet: "fd00:10:244::/48"
  serviceSubnet: "fd00:10:96::/112"
```

Deploy the cluster with `kind create cluster --config=kind-ipv6-only.yaml` and check that the nodes only receive an IPv6 address.

```
$ kubectl get nodes -o wide
NAME                 STATUS   ROLES           AGE   VERSION   INTERNAL-IP
kind-control-plane   Ready    control-plane   29m   v1.31.0
fc00:f853:ccd:e793::4
kind-worker          Ready    <none>          29m   v1.31.0
fc00:f853:ccd:e793::2
kind-worker2         Ready    <none>          29m   v1.31.0
fc00:f853:ccd:e793::3
```

Only IPv6 PodCIDRs are assigned to the node.

```
$ kubectl describe node kind-worker
[...]
PodCIDR:                      fd00:10:244:2::/64
PodCIDRs:                     fd00:10:244:2::/64
[...]
```

As IPv4 is enabled by default in Cilium, make sure to disable it during the Cilium installation. As mentioned earlier, as encapsulation mode is not supported, we'll use native routing mode and autoDirectNodeRoutes instead (these features will be explained in Chapter 5).

```
$ helm install cilium cilium/cilium \
  --version 1.17.1 \
  --namespace kube-system \
  --set ipv6.enabled=true \
  --set ipv4.enabled=false \
  --set ipam.mode=kubernetes \
  --set routingMode=native \
  --set autoDirectNodeRoutes=true \
  --set ipv6NativeRoutingCIDR=fd00:10:244::/48
```

Let's deploy the two pods once again.

```
$ kubectl apply -f pod1.yaml -f pod2.yaml
pod/pod-worker created
pod/pod-worker2 created
```

They have only been assigned an IPv6 address.

```
$ kubectl get pods -o wide
NAME          READY   AGE   IP
pod-worker    1/1     10m   fd00:10:244:2::4647
pod-worker2   1/1     10m   fd00:10:244:1::e1c
```

Let's also consider the service - notice a slight change on the ipFamilyPolicy and ipFamilies settings required to deploy IPv6-only services.

```
apiVersion: v1
kind: Service
metadata:
  name: echoserver
spec:
  ipFamilyPolicy: SingleStack
```

```
  ipFamilies:
  - IPv6
  ports:
    - port: 80
      targetPort: 80
      protocol: TCP
  type: ClusterIP
  selector:
    app: echoserver
```

Let's deploy the service and the associated deployment.

```
$ kubectl apply -f echo-kube-ipv6.yaml
deployment.apps/echoserver created
service/echoserver created
```

Pod-to-pod connectivity tests are successful.

```
$ IPv6=$(kubectl get pod pod-worker2 -o jsonpath='{.status.podIPs[0].ip}')
$ echo $IPv6
fd00:10:244:1::e1c
$ kubectl exec -it pod-worker -- ping6 -c 5 $IPv6
PING fd00:10:244:1::e1c (fd00:10:244:1::e1c) 56 data bytes
64 bytes from fd00:10:244:1::e1c: icmp_seq=1 ttl=60 time=0.179 ms
64 bytes from fd00:10:244:1::e1c: icmp_seq=2 ttl=60 time=0.090 ms
64 bytes from fd00:10:244:1::e1c: icmp_seq=3 ttl=60 time=0.094 ms
^C
--- fd00:10:244:1::e1c ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2039ms
rtt min/avg/max/mdev = 0.090/0.121/0.179/0.041 ms
```

So are pod-to-service and DNS resolution.

```
$ ServiceIPv6=$(kubectl get svc echoserver -o jsonpath='{.spec.clusterIP}')
$ echo $ServiceIPv6
fd00:10:96::5876
$
$ kubectl exec -i -t pod-worker -- curl -6 -o /dev/null -s -w "%{http_code}\n"
http://[$ServiceIPv6]/
200

$ kubectl exec -i -t pod-worker -- nslookup -q=AAAA echoserver.default
Server:         fd00:10:96::a
Address:        fd00:10:96::a#53

Name:   echoserver.default.svc.cluster.local
Address: fd00:10:96::5876

$ kubectl exec -i -t pod-worker -- curl -6 -o /dev/null -s -w "%{http_code}\n"
http://echoserver.default.svc
200
```

Conclusion

This chapter laid the foundation for Cilium networking by covering IP address management (IPAM) and IPv6/dual-stack support. We explored the various IPAM modes Cilium offers, from simple host-scope setups to more flexible options like cluster-scope and multi-pool, and discussed how to deploy clusters using IPv4, IPv6, or both.

These decisions are critical; changing them later can be disruptive, unsupported, or simply impractical. Our goal is to help you make informed, durable choices that align with your environment's long-term needs.

Now that you understand how Cilium assigns IP addresses to pods, the next step is to understand how traffic actually flows between them. In the next chapter, we'll dive into how packets are forwarded within and across nodes, examine the role of encapsulation, and compare native routing with overlay tunnels. We'll even dissect packet headers to see how Cilium, and the underlying network, moves traffic across your cluster.

About the Authors

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